RELEVANT EXTRACTION AND CONTRACTION OF LOCAL INTERACTIONS

Abstract. We study local Grassmann interactions on the three-dimensional chamber lattice, with independent ordinary and covariant-difference generators. Under explicit small-field and charge bounds, vacuum and scalar mass extraction followed by a shortest-chain adjustment makes the full ordinary quadratic kernel vanish at zero connection on small polymers. Scaling and reblocking give an explicit norm bound that is contractive for sufficiently large block scale and sufficiently small charge. The low-degree gain follows from local gauge representatives with an analytic disk of radius $L^{7/4}$. We also prove a local reconstruction estimate, uniform in the lattice mesh, using cancellation in the averaged chamber Dirac operator.

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1. Introduction

Let $L \geq 5$ be odd, let $M = L^m$ with $m \in \mathbb{Z}_{\geq 2}$, and put $\eta_k = L^{-k}$, $k \in \mathbb{Z}_{\geq 0}$. We use the photon torus $\eta_k \mathbb{Z}^3/\ell\mathbb{Z}^3$ and its chamber sites (c, g), where g is one of the 48 signed permutation matrices. Spinors have four components and are antiperiodic in each coordinate. A polymer is a face-connected union of the partition M-cubes. Its size parameter $Md_M(X)$ is the infimum of the lengths of connected finite nets meeting every cube of X; it is called small if $d_M(X) \leq L$. The net convention and its estimates are given in Appendix B.

An activity is a family of Grassmann polynomials in independent ordinary generators $\psi, \bar{\psi}$ and ordered-pair generators $\chi, \bar{\chi}$. The latter represent covariant Hölder differences only when the fields are evaluated. We work with the space $K_k^{x,\mathbb{C}}(\ell)$ defined in Section 2: its families are balanced, holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, local in the stated ordinary and pair indices and in the connection bonds, and covariant under the specified lattice, independent real-gauge and charge actions. The real subspace K_k^x satisfies the additional antilinear identity there. The two generator weights and family norm are

$$h_{k,1} = e_k^{-1/4}, \qquad h_{k,2} = e_k^{-1/4+\varepsilon}, \qquad \|\mathscr{E}\|_k^x = \sup_X e^{\kappa d_M(X)} \sup_{\widetilde{R}_k \times \mathfrak{X}_k} \|\mathscr{E}(X)\|_{h_k}.$$

Here $0 < e_k < 1$; the weighted coefficient norm and the two connection domains are specified in (5) and (3). There is no smallness assumption on the activity in the following linear theorem.

Set

$$\bar\varepsilon=\frac{1}{200},\quad c_X=54,\quad \kappa_0=54\log 72,\quad K_0=\frac{e^{\kappa_0}}{36},\quad c_{\mathrm{rb}}=c_XK_0,\quad c_{\mathrm{fl}}=61c_{\mathrm{rb}}.$$

Theorem A (Extraction and normalized contraction)**.** Assume that both $\ell/(LM)$ and $\ell/(L^2M)$ are odd integers at least 13. Suppose

$$\begin{gathered}0<\varepsilon\le\bar\varepsilon,\qquad 1-\varepsilon<\alpha<1,\qquad L^{1/8-3\bar\varepsilon/2}\ge 2400,\qquad e_k^{\varepsilon}\le 1/4,\\ 160L^{1+3\varepsilon/2}Me_k^{\varepsilon}\le 1,\qquad 9ML^{\varepsilon}e_k^{2\varepsilon}\le 1,\qquad 90c_XLMe_k^{\varepsilon}\le 1,\end{gathered}\tag{1}$$

and

$$\kappa\ge 7\kappa_0+7,\qquad L(\kappa-6\kappa_0-4)-1\ge\kappa.\tag{2}$$

Let $\mathscr{E}\in K_k^{x,\mathbb{C}}(\ell)$ and $e_{k+1}=L^{1/2}e_k$.

(i) There are complex-linear vacuum and mass maps, real on K_k^x, with

$$|\varepsilon_{\mathrm{v}}(\mathscr{E})|\le K_0M^{-3}\|\mathscr{E}\|_k^x,\qquad |m(\mathscr{E})|\le K_0M^{-3}e_k^{1/2}\|\mathscr{E}\|_k^x.$$

The extracted mass matrix is $m(\mathscr{E})1_4$. The local subtraction R of (17) satisfies $\|R\mathscr{E}\|_k^x\le 2\|\mathscr{E}\|_k^x$ and the exact independent-generator identity

$$\sum_X\mathscr{E}(X)=-\varepsilon_{\mathrm{v}}(\mathscr{E})\ell^3-m(\mathscr{E})\langle\bar\psi,\psi\rangle+\sum_X(R\mathscr{E})(X).$$

On a small polymer its zero-field vacuum kernel and the integral of its zero-field ordinary quadratic kernel vanish.

(ii) The shortest-chain adjustment of (23) gives $E=(R\mathscr{E})^{\mathrm{nat}}\in K_k^{x,\mathbb{C}}(\ell)$, with $\|E\|_k^x\le 20\|\mathscr{E}\|_k^x$. It leaves the evaluated value of $R\mathscr{E}$ unchanged whenever $\chi=\delta_{\alpha,\mathscr{A}+\mathscr{C}}\psi$ and $\bar\chi=\delta_{\alpha,-(\mathscr{A}+\mathscr{C})}\bar\psi$. On every small polymer it has the stronger normalization

$$E_{00}(X;0,0)=0,\qquad E_{20}(X;0,0;\cdot,\cdot)=0$$

as full antisymmetric kernels. Both subtraction and adjustment leave large-polymer activities unchanged.

(iii) Scale connections by $A_L(b)=L^{-1/2}A(b/L)$, ordinary generators by L^{-1}, and pair generators by $L^{-1-\alpha}$. Let $\bar X^L$ be the union of LM-cubes containing X, and set

$$(\mathcal{L}E)(Y;\mathscr{A},\mathscr{C})=\sum_{\bar X^L=LY}S_LE(X;\mathscr{A}_L,\mathscr{C}_L).$$

Then $\mathscr{E}\mapsto\mathcal{L}(R\mathscr{E})^{\mathrm{nat}}$ is a linear map into $K_{k+1}^{x,\mathbb{C}}(\ell/L)$ and

$$\|\mathcal{L}E\|_{k+1}^x\le c_{\mathrm{fl}}L^{-1/4+3\varepsilon/2}\|\mathscr{E}\|_k^x.$$

For $w_j^{\mathrm{E}}=e_j^{1/4-7\varepsilon}$, the relative estimate is

$$\frac{\|\mathcal{L}E\|_{k+1}^x}{w_{k+1}^{\mathrm{E}}}\le c_{\mathrm{fl}}L^{-3/8+11\varepsilon/2}\frac{\|\mathscr{E}\|_k^x}{w_k^{\mathrm{E}}}.$$

Its factor is at most $1/16$ if, in addition, $c_{\mathrm{fl}}L^{-3/8+6\bar\varepsilon}\le 1/16$.

All the stated maps preserve the real subspaces and commute with setting the second connection argument to zero. ■□

The proof reproduces the extraction, adjustment and contraction arguments of [Cur, §12]. The first obstruction is the ordinary quadratic term. Subtracting a local mass makes its integral vanish, but does not make its kernel vanish at each pair of points. We transport both legs along shortest chains of cubes and average their terminal points. The remaining ordinary quadratic term then factors through the integral of its kernel. The other terms contain difference generators, whose

scaling is stronger. This construction also explains why the independent generators are retained until evaluation.

The two lowest degrees require more than degree scaling. On a small polymer, a real quadratic gauge removes the real constant and symmetric linear parts of a scaled connection. Two cutoffs, one for the Taylor remainder and one for the residual linear field, produce global representatives in an analytic disk of radius $L^{7/4}$. The constant complex fields left over are removed by holomorphic continuation of real-gauge identities. The zero vacuum value and charge evenness give a second-order Cauchy gain; the full quadratic normalization gives a first-order gain. After reblocking, these estimates yield Theorem A. Sections 3–6 give the proof.

Local reconstruction is a separate estimate, used when unit-scale mean-field generators are replaced by ordinary and boundary-pair legs. Theorem B, stated with its own hypotheses in Section 7, gives uniform row bounds for that substitution. The apparent η_k^{-1} loss in the Dirac stencil cancels before estimation. The remaining phase defect is summed according to the first scale at which two neighboring sites have the same block center. Recovery of a mean field additionally assumes the fine operator is invertible at the connection in question. No inverse is needed for the row estimates or for Theorem A.

2. Activities, connections and coefficient norms

We specify the local activity space used in Theorem A. The conventions are those of [Cur, §12.1]. Only the finite Grassmann algebra, connection domains and symmetry actions enter the extraction and contraction arguments.

2.1. **Chamber sites, pairs and connection domains.** Write O_{cub} for the signed permutation matrices on $\mathbb{R}^3$. A chamber site is $x = (c, g)$, with photon projection $\pi x = c$ and $g \in O_{\mathrm{cub}}$. Integration over chamber sites uses $\mathrm{d}x = (\eta_k^3/48)\sum_{c,g}$. We take M-cubes centered at $M\mathbb{Z}^3/\ell\mathbb{Z}^3$; their lattice sites form a partition. Blocks of every odd side used below are centered in the same way.

For the three internal neighbors let s_0 change the sign of the third coordinate, let s_1 exchange the second and third coordinates, and let s_2 exchange the first and second. Let r_{e_1} change the sign of the first coordinate. Set

$$x^{(i)} = (c, gs_i) \quad (0 \le i \le 2), \qquad x^{(3)} = (c + \eta_k g e_1, g r_{e_1}).$$

Each neighbor map is an involution. An oriented bond field B is antisymmetric under reversal and periodic on the photon lattice. Write $B_\mu(c) = B(c, c + \eta_k e_\mu)$, and use the forward differences

$$\partial_\nu B_\mu(c) = \eta_k^{-1}(B_\mu(c + \eta_k e_\nu) - B_\mu(c)), \qquad (d\lambda)_\mu(c) = \eta_k^{-1}(\lambda(c + \eta_k e_\mu) - \lambda(c)).$$

Thus $(dB)_{\nu\mu} = \partial_\nu B_\mu - \partial_\mu B_\nu$. Distances between photon sites are sup distances, using the shortest lift when a distance bound is imposed. For $0 < \alpha < 1$ the three norms are

$$\|B\|_1 = \|B\|_\infty, \qquad \|B\|_2 = \|\partial B\|_\infty, \qquad \|B\|_3 = \|\delta_\alpha \partial B\|_\infty = \sup_{\substack{\mu,\nu,c,c' \\ 0<d(c,c')\le 1}} \frac{|\partial_\nu B_\mu(c') - \partial_\nu B_\mu(c)|}{d(c,c')^\alpha}.$$

Local versions use only the indicated cells in the given region. Let $\square^\dagger$ be the concentric enlargement of an M-cube to side $(2L+1)M$. Put $\theta_i = e_k^{-3/4+i\varepsilon}$. The connection domains are

$$\begin{aligned} R_k &= \{B : \|B\|_i < \theta_i,\ 1 \le i \le 3\}, \\ \widetilde{R}_k &= \{B_0 + B_1 : B_1 \in R_k,\ B_0 \text{ real, and on each } \square^\dagger, \\ &\qquad\qquad B_0 = B_\square^{\mathrm{b}} + d\lambda_\square \text{ with } B_\square^{\mathrm{b}}, \lambda_\square \text{ real and } \|B_\square^{\mathrm{b}}\|_i < \theta_i\}, \\ \mathfrak{X}_k &= \{C : \|C\|_\infty < e_k^{-1+2\varepsilon},\ \|\partial C\|_\infty < e_k^{-1+3\varepsilon}\}. \end{aligned} \tag{3}$$

Only existence of these representatives is required. For $0 < c \le 1$, the notation $c\widetilde{R}_k$ multiplies all three bounds for both summands by c. For $\widetilde{R}_k^-$ the local real bounds remain θ_i, while the global summand has bounds $(1 - e_k^\varepsilon)\theta_i$.

For a short ordered pair $x = (c, g), x' = (c', g')$, let $\tau B(c, c')$ be the arithmetic mean of the sums of B along the six coordinate-order paths from c to c'. The bond values in this sum are unscaled; define

$$U_B(x, x') = \exp\{ie_k\eta_k\tau B(c, c')\}, \qquad (\delta_{\alpha,B}\psi)(x, x') = \frac{U_B(x, x')\psi(x') - \psi(x)}{\max(|c - c'|_\infty, \eta_k)^\alpha}.$$

For pairs within one cluster the transport is one. Barred fields use U_B^{-1}. These formulas specify evaluation; the pair generators in the algebra remain independent.

We use Pauli matrices and

$$\gamma_\mu = \sigma_3 \otimes \sigma_\mu, \quad \gamma_5 = \sigma_1 \otimes 1_2, \quad \Sigma_3 = \sigma_3 \otimes 1_2, \qquad \mathsf{C} = i\gamma_2, \quad \varkappa = \gamma_5\mathsf{C} = \sigma_2 \otimes \sigma_2.$$

Here C is real with $\mathsf{C}^T = \mathsf{C}^{-1} = -\mathsf{C}$, while $\varkappa$ is a real symmetric involution. The spin group below is the group generated by $\gamma_1, \gamma_2, \gamma_3, \gamma_5$. Its diagonal coordinate action $r(S)$ is determined by $S\gamma_\mu S^{-1} = r(S)_{\mu\mu}\gamma_\mu$. Connections transform by pullback on oriented bonds.

2.2. **The local Grassmann algebra.** Fix a level k, mesh $\eta_k = L^{-k}$ and an admissible side ℓ, with $\ell/(LM)$ odd and at least 13. Let $\mathcal{P}_k$ be the collection of face-connected polymers of M-cubes on this torus. A polymer is small when $d_M(X) \le L$. A small polymer has an injective lift, unique up to a period translation, by the diameter and lifting estimates of Appendix B. Write $\mathrm{Vol}(X) = M^3|X|_M$.

The independent Grassmann generators are ψ, $\bar\psi$ on the chamber sites $x = (c, g)$ and χ, $\bar\chi$ on ordered pairs. For distinct chamber sites on the covering lattice, put $d_X(x, x') = \max(|c - c'|_\infty, \eta_k)$ and restrict pairs to $d_X \le M$. Pairs on the torus are simultaneous period classes; their shortest lifts are unique since $2M < \ell$. A pair is admissible for X when its photon sites and all six rectilinear paths between them lie in X, or when it is $(x, x^{(3)})$ and one of its photon sites belongs to X. Thus a pair may leave X only along a bond of type 3. Pairs inside a cluster have $d_X = \eta_k$ and are admissible when their common photon site belongs to X. Ordinary generators acquire the sign $(-1)^{n_1+n_2+n_3}$ under translation by ℓn. Pair generators acquire the same sign when both endpoints are translated by ℓn. The kernels are antiperiodic in each ordinary argument and in each simultaneous pair argument with these conventions.

Let $\xi = (x, \beta, \omega)$ index an ordinary leg and $\zeta = (x, x', \beta, \omega)$ a Hölder leg, where $\omega = 0$ is unbarred and $\omega = 1$ is barred. We use the index measures

$$\nu(\xi) = \eta_k^3/48, \qquad \nu(\zeta) = (\eta_k^3/48)^2.$$

Integrals sum the spin and bar indices as well. An activity has a unique expansion with separately antisymmetric kernels,

$$\mathscr{E}(X; \mathscr{A}, \mathscr{C}) = \sum_{n,m} \frac{1}{n!m!} \int \mathrm{d}\xi\, \mathrm{d}\zeta\, \mathscr{E}_{nm}(X; \mathscr{A}, \mathscr{C}; \xi; \zeta)\psi(\xi_1)\cdots\psi(\xi_n)\chi(\zeta_1)\cdots\chi(\zeta_m). \tag{4}$$

Barred generators are included in this notation. The generator weights, as distinct from the index measures, are

$$h_{k,1} = e_k^{-1/4}, \qquad h_{k,2} = e_k^{-1/4+\varepsilon}.$$

For an element F the weighted coefficient norm is

$$\|F\|_{h_k} = \sum_{n,m} \frac{h_{k,1}^n h_{k,2}^m}{n!m!} \int \mathrm{d}\xi\, \mathrm{d}\zeta\, |F_{nm}|. \tag{5}$$

In particular a generator has norm $h_{k,i}$; the weight ν enters only when coefficients are written as integral kernels.

An activity family in $K_k^{x,\mathbb{C}}$ is balanced: every monomial has as many barred as unbarred legs. Its kernels are holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, vanish unless all ordinary legs lie over X and all pairs are admissible for X, and read the two connections only on bonds meeting X. The norm is

$$\|\mathscr{E}\|_k^x = \sup_{X\in\mathcal{P}_k} \exp(\kappa d_M(X)) \sup_{(\mathscr{A},\mathscr{C})\in\widetilde{R}_k\times\mathfrak{X}_k} \|\mathscr{E}(X;\mathscr{A},\mathscr{C})\|_{h_k}. \tag{6}$$

We require $\kappa \geq \kappa_0$ and finite norm.

The required symmetries act on both the connections and the generators. For $\tilde{g} = (S, a)$ with the coordinate sign action $r(S)$ and $a \in M\mathbb{Z}^3$, put

$$\tilde{g}(c, g) = (r(S)c + a, r(S)g), \qquad a_{\tilde{g}}\psi(x) = S\psi(\tilde{g}^{-1}x), \qquad a_{\tilde{g}}\bar{\psi}(x) = (S^{-1})^T\bar{\psi}(\tilde{g}^{-1}x).$$

The same spin matrices act on pair generators after transforming both endpoints. Local gauge transformations multiply an unbarred ordinary or pair leg by $\exp(ie_k\Lambda(c))$, using the photon site of its first point, and a barred leg by the inverse phase. Write this automorphism as G_Λ. Charge conjugation and the antilinear reality map act by

$$\mathrm{Cg}\,\psi = \mathsf{C}\bar{\psi}, \qquad \mathrm{Cg}\,\bar{\psi} = -\mathsf{C}\psi, \qquad \mathbf{K}\psi = \varkappa\psi, \qquad \mathbf{K}\bar{\psi} = \varkappa\bar{\psi},$$

and by the same rules on the Hölder generators.

The covariance requirements on a family are

$$\begin{aligned} a_{\tilde{g}}\mathscr{E}(\tilde{g}X;\tilde{g}\mathscr{A},\tilde{g}\mathscr{C}) &= \mathscr{E}(X;\mathscr{A},\mathscr{C}),\\ \mathrm{Cg}\,\mathscr{E}(X;-\mathscr{A},-\mathscr{C}) &= \mathscr{E}(X;\mathscr{A},\mathscr{C}). \end{aligned} \tag{7}$$

For real scalar functions λ, λ' on the photon torus and two connection pairs in the domain which satisfy $\mathscr{A}^1 = \mathscr{A} - d\lambda$ and $\mathscr{C}^1 = \mathscr{C} - d\lambda'$ on bonds at distance at most one from X, the local gauge requirement is

$$G_{\lambda+\lambda'}\mathscr{E}(X;\mathscr{A}^1,\mathscr{C}^1) = \mathscr{E}(X;\mathscr{A},\mathscr{C}).$$

The real subspace K_k^x has in addition

$$\mathbf{K}\mathscr{E}(X;-\overline{\mathscr{A}},-\overline{\mathscr{C}}) = \mathscr{E}(X;\mathscr{A},\mathscr{C}).$$

All requirements refer to the generators before evaluating fields. They are essential to the form of the counterterms.

The spaces $K_k^{\mathbb{C}}, K_k$ without the second connection argument have the same definitions on $\widetilde{R}_k$. Restriction $\mathscr{E}|_0(X;\mathscr{A}) = \mathscr{E}(X;\mathscr{A},0)$ maps $K_k^{x,\mathbb{C}}$ into $K_k^{\mathbb{C}}$ and has norm at most one. The real subspaces also restrict. Analyticity, locality and the symmetries follow by setting the second argument and its gauge function to zero; the norm bound follows from the smaller supremum.

2.3. **Total connections.** The comparison domain $\widetilde{G}_k$ consists of fields $B_0 + B_1$ with B_0 real and $\|B_1\|_\infty < e_k^{-1+\varepsilon}$, where on each $\square^{\sim 5}$ the field B_0 has a real-gauge representative of supremum norm below $e_k^{-1+\varepsilon}$. Here $\square^{\sim 5}$ has the same center and side $11M$. The real-gauge representatives in the domains defined above may be chosen independently on each enlarged cube.

Assume $L \geq 5$, $0 < \varepsilon \leq 1/200$ and

$$e_k^\varepsilon \leq \tfrac{1}{4}, \qquad e_k^{1/4} + e_k^\varepsilon \leq 1, \qquad 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \leq 1. \tag{8}$$

Then

$$\widetilde{R}_k + \mathfrak{X}_k \subset \widetilde{G}_k. \tag{9}$$

Indeed, write $\mathscr{A} = A_0 + A_1$ with $A_1 \in R_k$ and A_0 locally real-gauge equivalent on $\square^\dagger$ to $A_\square^b$ with the three threshold bounds. Restrict the representative to $\square^{\sim 5} \subset \square^\dagger$. Its supremum norm is below $e_k^{-3/4+\varepsilon} \leq e_k^{-1+\varepsilon}$, and

$$\|A_1 + \mathscr{C}\|_\infty < e_k^{-3/4+\varepsilon} + e_k^{-1+2\varepsilon} = (e_k^{1/4} + e_k^\varepsilon)e_k^{-1+\varepsilon} \leq e_k^{-1+\varepsilon}.$$

This is the required decomposition of the total connection. Real gauge transformations can absorb any real constant part, locally on the nonwrapping cubes.

Put $\mathfrak{i}(B) = e_k \| \operatorname{Im} B\|_\infty$ and $q_d(B) = e_k \|dB\|_\infty$. The same decomposition gives

$$\mathfrak{i}(\mathscr{A}+\mathscr{C}) \le e_k^{1/4+\varepsilon} + e_k^{2\varepsilon} \le 1, \qquad q_d(\mathscr{A}+\mathscr{C}) \le 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \le 1. \tag{10}$$

For the curvature, every plaquette lies in an enlarged cube, and $dA_0 = dA^b_\square$ there. This costs at most $2\|\partial A^b_\square\|_\infty$; the global small field costs another $2\|\partial A_1\|_\infty$, and the external field costs $2\|\partial \mathscr{C}\|_\infty$. The imaginary bound uses only that A_0 is real.

The domains are open: the strict inequalities define open sets for the global summand, which can absorb a sufficiently small complex perturbation while the local real representative is kept fixed. The norm-defined sets are invariant under sign and conjugation; the local real-gauge definitions are also invariant under real gauges and real constant additions. Coordinate sign changes and M translations preserve them. Their derivative and Hölder norms are preserved, with a possible reversal of a bond and translation of its base point. Thus (9) applies also at the signs, conjugates and real gauges used below.

2.4. **Weighted linear substitution.** We will use the following finite-algebra fact, namely the linear-substitution part of [Cur, Proposition 11.4]. Positive index measures do not change it: when a polynomial is written in ordered monomials, each coefficient already includes the product of its index measures.

Lemma 2.1. Give each generator of a finite Grassmann algebra a positive weight, and give the algebra the sum of absolute ordered coefficients times the corresponding products of weights. This norm is submultiplicative. A homomorphism that sends each generator to an odd linear combination of target generators with norm at most its original weight decreases this norm. ■

Proof. In multiplying two ordered monomials, a repeated generator makes the product zero. Otherwise the product differs from the ordered union only by a sign, and its weight is the product of the two weights. The triangle inequality, summed over the two sets of coefficients, proves submultiplicativity. The norm of the image of each ordered monomial is consequently at most its original weight. Summing its absolute coefficient proves the substitution bound. Separately antisymmetric kernels give the same coefficient sum: the factorials cancel the number of orders of each set of distinct indices. □

3. Extraction and quadratic normalization

The extraction and shortest-chain adjustment follow [Cur, §§12.3, 12.5]. They solve different problems. The extracted matrix becomes scalar only after summing over polymers. The adjustment then converts the vanishing integral of each remaining quadratic kernel into vanishing of the full kernel at zero connection. This is the normalization used in the Cauchy estimate.

3.1. **Vacuum and mass extraction.** For a balanced activity, its part of bidegree $(2,0)$ can be written uniquely as

$$\mathscr{E}^{[2,0]}(X;\mathscr{A},\mathscr{C}) = \int \mathrm{d}x\, \mathrm{d}x'\, \bar\psi(x)^T K_X(\mathscr{A},\mathscr{C};x,x')\psi(x').$$

The matrix kernel is the ordinary antisymmetric kernel with the first bar index one and the second bar index zero. Antisymmetry accounts for the other ordering; in particular

$$\|\mathscr{E}^{[2,0]}(X;\mathscr{A},\mathscr{C})\|_{h_k} = h_{k,1}^2 \sum_{\beta,\beta'} \int \mathrm{d}x\, \mathrm{d}x' |K_X(\mathscr{A},\mathscr{C};x,x')_{\beta\beta'}|. \tag{11}$$

The kernel is antiperiodic in each argument separately.

For a small X, choose one lift $\widetilde{X}$ and define

$$a_0(\mathscr{E},X)=\mathrm{Vol}(X)^{-1}\mathscr{E}_{00}(X;0,0),\qquad a_2(\mathscr{E},X)=\mathrm{Vol}(X)^{-1}\int_{\widetilde{X}}\mathrm{d}x\int_{\widetilde{X}}\mathrm{d}x'\,K_X(0,0;x,x'). \tag{12}$$

Both cluster labels are averaged with measure $\eta_k^3/48$. The second expression is independent of the lift: simultaneous translation by a period multiplies its two arguments by the same antiperiodic sign, whose product is one.

For an M-cube $\square$ put

$$\varepsilon_{\mathrm{v}}(\mathscr{E},\square)=-\sum_{\substack{X\supset\square\\ X\text{ small}}}a_0(\mathscr{E},X),\qquad M(\mathscr{E},\square)=-\sum_{\substack{X\supset\square\\ X\text{ small}}}a_2(\mathscr{E},X). \tag{13}$$

Translations by $M\mathbb{Z}^3$ identify the small-polymer sets over different cubes. They leave the two extracted quantities unchanged. Thus the two sums do not depend on $\square$. Write their values as $\varepsilon_{\mathrm{v}}(\mathscr{E})$ and $M(\mathscr{E})$.

Proposition 3.1. There is a complex-linear scalar map $m(\mathscr{E})$ such that $M(\mathscr{E})=m(\mathscr{E})1$. Both m and ε_{v} are real on K_k^x, and

$$|\varepsilon_{\mathrm{v}}(\mathscr{E})|\le K_0M^{-3}\|\mathscr{E}\|_k^x,\qquad |m(\mathscr{E})|\le K_0M^{-3}e_k^{1/2}\|\mathscr{E}\|_k^x. \tag{14}$$

They commute with restriction of the second connection argument to zero. ■

Proof. Substitution of the generator transformations in a bilinear, followed by uniqueness of its kernel, gives

$$\begin{aligned}a_2(\mathscr{E},\tilde{g}X)&=Sa_2(\mathscr{E},X)S^{-1},\\ a_2(\mathscr{E},X)&=\mathsf{C}^{-1}a_2(\mathscr{E},X)^T\mathsf{C},\\ a_2(\mathscr{E},X)&=\varkappa^{-1}\overline{a_2(\mathscr{E},X)}\varkappa\quad(\mathscr{E}\in K_k^x).\end{aligned} \tag{15}$$

For the first relation, the two variables and their labels are transformed together, and the measure is unchanged. For charge conjugation, moving the unbarred leg past the barred leg cancels the minus sign in its transformation, and exchanges the two kernel variables. For the last relation, the reality map conjugates the coefficient and multiplies both legs by $\varkappa$; its inverse transpose is $\varkappa^{-1}$. The constant kernel has the corresponding scalar invariance and is real for a real family.

Let z be the center of $\square$ and take $S=\gamma_1,\gamma_2,\gamma_3$ or γ_5. The symmetry $(S,(1-r(S))z)$ fixes $\square$ and permutes the small polymers containing it. The first relation in (15) therefore shows that $M(\mathscr{E})$ commutes with all four matrices. Their commutant is scalar. In the fixed Dirac basis, γ_3 and $\Sigma_3=-i\gamma_1\gamma_2\gamma_3$ are diagonal with the four distinct pairs of diagonal entries $(1,1),(-1,1),(-1,-1),(1,-1)$. A commuting matrix is diagonal; commutation with γ_1 identifies its first two and last two entries, and commutation with γ_5 identifies these pairs. This proves the scalar assertion. The last relation in (15) makes this scalar real for a real family.

The coefficient norm gives, for each small X,

$$\begin{aligned}\mathrm{Vol}(X)|a_0(\mathscr{E},X)|&\le\|\mathscr{E}(X;0,0)\|_{h_k},\\ \mathrm{Vol}(X)\sum_{\beta,\beta'}|a_2(\mathscr{E},X)_{\beta\beta'}|&\le h_{k,1}^{-2}\|\mathscr{E}^{[2,0]}(X;0,0)\|_{h_k}.\end{aligned} \tag{16}$$

Use $h_{k,1}^{-2}=e_k^{1/2}$ and $\mathrm{Vol}(X)\ge M^3$. Each contribution is bounded by the corresponding right side of (14), times $\exp(-\kappa d_M(X))$. The polymer sum over a fixed cube is at most K_0 by $\kappa\ge\kappa_0$. For the mass use one diagonal entry of the matrix sum, now known to be scalar. This proves the bounds. Linearity follows from uniqueness of the antisymmetric kernels. Restriction gives the same kernels at $(0,0)$, proving the last assertion. □

Subtract these local terms only on small polymers:

$$(R\mathscr{E})(X;\mathscr{A},\mathscr{C}) = \mathscr{E}(X;\mathscr{A},\mathscr{C}) - a_0(\mathscr{E},X)\operatorname{Vol}(X) - \int_X \mathrm{d}x\, \bar{\psi}(x)^T a_2(\mathscr{E},X)\psi(x). \tag{17}$$

For large X set $R\mathscr{E} = \mathscr{E}$. Then

$$\sum_X \mathscr{E}(X;\mathscr{A},\mathscr{C}) = -\varepsilon_{\mathrm{v}}(\mathscr{E})\ell^3 - m(\mathscr{E})\langle\bar{\psi},\psi\rangle + \sum_X (R\mathscr{E})(X;\mathscr{A},\mathscr{C}). \tag{18}$$

Indeed, decompose the integral and volume of each small polymer over its M-cubes, interchange the finite sums, and use (13) on each cube. Every chamber component receives the same scalar mass coefficient:

$$m(\mathscr{E})\langle\bar{\psi},\psi\rangle = m(\mathscr{E})\frac{\eta_k^3}{48}\sum_c\sum_g\sum_{\beta=1}^{4}\bar{\psi}_\beta(c,g)\psi_\beta(c,g).$$

The diagonal bilinear of a constant matrix A on X has kernel $48\eta_k^{-3}1[x = x' \text{ over } X]A$ when both arguments are taken over the same lift. Its integral is $\operatorname{Vol}(X)A$, and its norm is

$$h_{k,1}^2 \operatorname{Vol}(X)\sum_{\beta,\beta'}|A_{\beta\beta'}|.$$

These identities and (16) show that

$$\begin{gathered}(R\mathscr{E})_{00}(X;0,0) = 0 \quad (X \text{ small}),\\ \int_{\widetilde{X}} \mathrm{d}x\,\mathrm{d}x'\,(R\mathscr{E})_{20}(X;0,0;x,x') = 0 \quad (X \text{ small}),\\ \|R\mathscr{E}\|_k^x \le 2\|\mathscr{E}\|_k^x.\end{gathered} \tag{19}$$

For the bound, the two subtracted terms have distinct bidegrees and together have norm at most $\|\mathscr{E}(X;0,0)\|_{h_k}$. Apply the triangle inequality to (17), then the activity norm.

The map R preserves the activity spaces and restriction to zero. The subtracted kernels are independent of the connections, and their two legs lie over X, so analyticity and locality remain. A diagonal bilinear is invariant under every local gauge. Its lattice, charge and reality transformations follow from (15); the volume and constant term have the same symmetries. This verifies each requirement on the remainder, without assuming that an individual matrix $a_2(\mathscr{E},X)$ is scalar.

For later comparisons, the extraction has a precise local meaning across different sides. Fix a lifted M-cube and consider all lifted small polymers containing it. Their projections are bijective onto the small polymers containing its image, and their d_M and volumes are unchanged. If two families have the same constant and quadratic kernels on every such lifted polymer at zero connections, then their vacuum and mass matrices over this cube agree term by term in (13). Volume independence of a flow will require this equality of kernels from the step comparison.

3.2. **Adjustment of the quadratic kernel.** The extraction makes the integral of the zero-field quadratic kernel vanish. For contraction we need the kernel itself to vanish. We achieve this by writing each ordinary leg as a transported average and a sum of Hölder differences. This changes the activity before evaluation and leaves its evaluated value unchanged.

Let X be small, choose a lift $\widetilde{X}$, and let $x = (c,g)$ and $\tilde{x} = (\tilde{c},\tilde{g})$ lie over this lift. Take all shortest face chains of its M-cubes between their two cubes. For a chain $(\square_0,\dots,\square_n)$ with centers z_i, use the points

$$p_0 = x, \qquad p_{i+1} = (z_i, g) \quad (0 \le i \le n), \qquad p_{n+2} = \tilde{x}.$$

The initial label is kept at each intermediate center. Each distinct consecutive pair is admissible for X of the first kind. The paths in the first and last cube remain in those cubes; between adjacent centers all six paths are the straight segment across their common face. The last pair can have different labels, and can lie inside one cluster. For a coincident pair there is no difference term.

With $u_B(p,p') = U_B(p,p')$ and $u_B(p,p) = 1$, set

$$U_i^{\pm} = \prod_{j<i} u_B(p_j, p_{j+1})^{\pm 1}.$$

The chain has no repeated cube, and $n+1 \le |X|_M \le c_X(L+1)$. Hence

$$\sum_{i=0}^{n+1} |\pi p_i - \pi p_{i+1}|_1 \le (n+3)M \le 3c_X LM, \qquad \sum_{i:p_i \neq p_{i+1}} d_X(p_i,p_{i+1})^{\alpha} \le 3c_X LM. \tag{20}$$

The second estimate uses $d_X^{\alpha} \le \max(1, d_X) \le M$. The transport bound is $|u_B(p,p')^{\pm 1}| \le \exp(e_k \| \operatorname{Im} B\|_{\infty} |\pi p - \pi p'|_1)$, obtained by counting the bonds in each path.

For $B = \mathscr{A} + \mathscr{C}$, the domain decomposition and $e_k^{\varepsilon} \le 1/4$ give $e_k \| \operatorname{Im} B\|_{\infty} \le e_k^{\varepsilon}/2$. Assume in addition

$$90 c_X LM e_k^{\varepsilon} \le 1. \tag{21}$$

The modulus of every partial product $U_i^{\pm}$ is then at most $\exp(1)$, uniformly in the chain and mesh.

Let Mn be the mean over the shortest chains. Define the homomorphism $\Theta_{X,B}$ on ordinary generators by

$$\begin{aligned}
\Theta_{X,B}\psi(x) &= \operatorname{Vol}(X)^{-1} \int_{\widetilde{X}} \mathrm{d}\tilde{x}\ \operatorname{Mn} \left[U_{n+2}^{+} \psi(\tilde{x}) - \sum_{i:p_i \neq p_{i+1}} U_i^{+} d_X(p_i,p_{i+1})^{\alpha} \chi(p_i,p_{i+1}) \right], \\
\Theta_{X,B}\bar{\psi}(x) &= \operatorname{Vol}(X)^{-1} \int_{\widetilde{X}} \mathrm{d}\tilde{x}\ \operatorname{Mn} \left[U_{n+2}^{-} \bar{\psi}(\tilde{x}) - \sum_{i:p_i \neq p_{i+1}} U_i^{-} d_X(p_i,p_{i+1})^{\alpha} \bar{\chi}(p_i,p_{i+1}) \right].
\end{aligned} \tag{22}$$

The spin components are unchanged. Both displayed images are linear combinations of odd independent generators, so they anticommute and determine a unique homomorphism on the subalgebra of ordinary legs over X. Translation of the lift by a period preserves every transport and chain and multiplies each generator by the same antiperiodic sign. Thus the homomorphism is independent of this choice of lift.

If the generators are evaluated at fields satisfying their covariant Hölder relations, repeated substitution of

$$\psi(p_i) = u_B(p_i,p_{i+1})\psi(p_{i+1}) - d_X(p_i,p_{i+1})^{\alpha}\chi(p_i,p_{i+1})$$

along a chain gives exactly the first expression in (22); the barred relation uses u_B^{-1}. Taking the two means therefore reproduces the original leg. This proves equality after evaluation for every polynomial in the ordinary legs over X.

The adjusted family is

$$\mathscr{E}^{\mathrm{nat}}(X) = \mathscr{E}(X) - \mathscr{E}^{[2,0]}(X) + \Theta_{X,\mathscr{A}+\mathscr{C}} \mathscr{E}^{[2,0]}(X) \qquad (X \text{ small}), \tag{23}$$

and is unchanged on large polymers. The equality just proved shows that the adjusted and original families have equal evaluated values. Only bidegrees $(2,0), (1,1), (0,2)$ can change.

At zero connection all $U_i^{\pm}$ are one. The ordinary part of an adjusted bilinear with kernel K is therefore

$$\bar{\psi}_{\mathrm{av}}^{T} \left[\int_{\widetilde{X}} \mathrm{d}x\, \mathrm{d}x'\, K(x,x') \right] \psi_{\mathrm{av}}, \qquad \psi_{\mathrm{av}} = \operatorname{Vol}(X)^{-1} \int_{\widetilde{X}} \mathrm{d}x\, \psi(x).$$

Apply this to $R\mathscr{E}$, whose kernel integral vanishes. With $E = (R\mathscr{E})^{\mathrm{nat}}$, we obtain the stronger normalization

$$E_{00}(X;0,0) = 0, \qquad E_{20}(X;0,0;\cdot,\cdot) = 0 \quad (X \text{ small}). \tag{24}$$

The same chain estimate controls the norm of adjustment. Decompose $\Theta\psi = L_\psi + L_\chi$ by bidegree. For arbitrary generator weights (w_1, w_2),

$$\|L_\psi\|_w \le \exp(1) w_1, \qquad \|L_\chi\|_w \le \exp(1)(3c_X LM) w_2.$$

The normalized site and label mean has total mass one. Writing $s_K = \sum_{\beta,\beta'} \int |K_{\beta\beta'}|$ for the balanced quadratic kernel and $F = \Theta\mathscr{E}^{[2,0]}$ gives the three bounds

$$\begin{aligned}\|F^{[2,0]}\|_w &\le \exp(2) w_1^2 s_K,\\ \|F^{[1,1]}\|_w &\le 2\exp(2)(3c_X LM) w_1 w_2 s_K,\\ \|F^{[0,2]}\|_w &\le \exp(2)(3c_X LM)^2 w_2^2 s_K.\end{aligned}$$

Since $h_{k,2} = e_k^{\varepsilon} h_{k,1}$ and $3c_X LM e_k^{\varepsilon} \le 1/30$, summing these bounds yields

$$\|\mathscr{E}^{\mathrm{nat}}(X)\|_{h_k} \le \exp(2)(1 + 3c_X LM e_k^{\varepsilon})^2 \|\mathscr{E}(X)\|_{h_k} \le 10\|\mathscr{E}(X)\|_{h_k}. \tag{25}$$

Subtracting the original quadratic part before adding its image uses additivity of the norm over bidegrees.

All new legs lie over X or at its admissible chain pairs, and all new transports read bonds with both ends in X. Thus adjustment preserves locality. It preserves analyticity because these transports are entire in the connection. For the lattice covariance, symmetries carry shortest chains, their centers and labels to the corresponding image chains. For a real gauge the factors along a partial chain telescope to the phase difference at its endpoints, giving the original leg phase. Charge conjugation replaces u_B by u_B^{-1}; the reality map uses $\overline{u_{-\overline{B}}} = u_B$. These identities on generators prove all four covariance identities for Θ, and hence the activity symmetries for adjustment. Restriction to $\mathscr{C} = 0$ commutes with adjustment because it leaves the same total connection and the same ordinary quadratic kernel. Therefore adjustment preserves the complex and real activity spaces.

4. Scaling and reblocking

Throughout Sections 4–6, both ℓ and ℓ/L are admissible and the hypotheses of Theorem A hold. In particular $\ell \ge 13L^2M$. We first record the exact scaling identities from [Cur, §12.6].

Write the three thresholds of R_k as

$$\theta_1 = e_k^{-3/4+\varepsilon}, \qquad \theta_2 = e_k^{\varepsilon}\theta_1, \qquad \theta_3 = e_k^{\varepsilon}\theta_2,$$

and the thresholds of $\mathfrak{X}_k$ as $d_1 = e_k^{-1+2\varepsilon}$ and $d_2 = e_k^{\varepsilon} d_1$. For a level $k+1$ bond function and gauge function define

$$A_L(b) = L^{-1/2} A(b/L), \qquad \lambda_L(c) = L^{1/2}\lambda(c/L). \tag{26}$$

Since $\eta_{k+1} = \eta_k/L$, the finite differences obey $(d\lambda)_L = d\lambda_L$ and

$$\|A_L\|_\infty = L^{-1/2}\|A\|_\infty, \qquad \|\partial A_L\|_\infty = L^{-3/2}\|\partial A\|_\infty, \qquad \|\delta_\alpha \partial A_L\|_\infty \le L^{-3/2-\alpha}\|\delta_\alpha \partial A\|_\infty.$$

The last inequality uses only the coarse pairs at distance $\le 1/L$, a subset of those in the coarse seminorm. The charge relation $e_{k+1} = L^{1/2} e_k$ therefore gives the following fine-scale bounds for the scaled complex small field, and for each scaled local real representative:

$$\begin{aligned} b_1 &= L^{-7/8+\varepsilon/2}\theta_1,\\ b_2 &= L^{-15/8+\varepsilon}\theta_2,\\ b_3 &= L^{-15/8+3\varepsilon/2}\theta_3.\end{aligned} \tag{27}$$

Here the additional factor $L^{-\alpha}$ in the third bound has been discarded. For the external argument the bounds are

$$\|\mathscr{C}_L\|_\infty < L^{-1+\varepsilon} d_1, \qquad \|\partial\mathscr{C}_L\|_\infty < L^{-2+3\varepsilon/2} d_2. \tag{28}$$

The enlarged coarse cube used to define a local representative, after multiplication by L, contains the enlarged fine cube. Scaling its real gauge thus gives the corresponding fine representative. All the ratios in (27) and (28) are at most $L^{-3/4}$, by (1). It follows that

$$\mathscr{A}_L \in L^{-3/4}\widetilde{R}_k \subset \widetilde{R}_k^-, \qquad \mathscr{C}_L \in L^{-3/4}\mathfrak{X}_k. \tag{29}$$

For the second inclusion in the first statement use $L^{-3/4} \le 1/2 \le 1 - e_k^{\varepsilon}$.

The Grassmann scaling homomorphism is

$$S_L\psi(x) = L^{-1}\psi'(x/L), \qquad S_L\chi(x,x') = L^{-1-\alpha}\chi'(x/L, x'/L), \tag{30}$$

with the same powers for the barred generators. Only the photon coordinate is divided by L; the chamber label and spin index are kept. Ordinary indices are mapped bijectively onto the coarse ordinary indices. Pair indices are mapped bijectively onto the coarse pair indices with $d_X \le M/L$, since d_X is divided by L. This includes a pair inside one cluster: $d_X = \eta_k$ becomes $d_X = \eta_{k+1}$. The antiperiodic signs agree under this bijection.

Thus distinct basis monomials have distinct images, up to their ordering signs, and every monomial of bidegree (n,m) is multiplied by $L^{-n-(1+\alpha)m}$. There is no cancellation of coefficients. For arbitrary positive weights,

$$\|S_L F\|_{w_1,w_2} = \|F\|_{L^{-1}w_1, L^{-1-\alpha}w_2}. \tag{31}$$

As $h_{k+1,1}/h_{k,1} = L^{-1/8}$ and $h_{k+1,2}/h_{k,2} = L^{-1/8+\varepsilon/2}$, this gives

$$\|(S_L F)^{[n,m]}\|_{h_{k+1}} = L^{-p(n,m)}\|F^{[n,m]}\|_{h_k}, \qquad p(n,m) = n + (1+\alpha)m + \frac{n+m}{8} - \frac{m\varepsilon}{2}. \tag{32}$$

The powers are identical for all admissible pair types.

In kernel notation the same calculation reads

$$(S_L F)_{nm}(\xi';\zeta') = \begin{cases} L^{3n+6m-n-(1+\alpha)m} F_{nm}(L\xi'; L\zeta'), & d_X(\zeta'_j) \le M/L \text{ for every } j, \\ 0, & \text{otherwise.} \end{cases} \tag{33}$$

Indeed, the index weights satisfy $\eta_k^3/48 = L^3\eta_{k+1}^3/48$ and hence contribute L^{3n+6m} to the kernel. Antisymmetry and antiperiodicity are preserved, so uniqueness of the kernel expansion gives (33).

Let $\bar{X}^L$ be the union of LM-cubes containing the cubes of X. Define the reblocked family by

$$(\mathcal{L}F)(Y;\mathscr{A},\mathscr{C}) = \sum_{X:\bar{X}^L = LY} S_L F(X;\mathscr{A}_L,\mathscr{C}_L). \tag{34}$$

The sum is finite on a fixed torus. It is local in Y if F is local in each X. Ordinary legs over X scale to legs over Y; the six paths for a pair of the first kind scale to paths in Y. For a boundary pair, $x^{(3)}/L = (x/L)^{(3)}$, and an endpoint in X scales to an endpoint in Y. Finally, every fine bond meeting X scales to a coarse bond meeting Y. Equality of both coarse connection arguments on these bonds consequently gives equality of the fine arguments read by each term. This proves both the support and the coefficient locality of (34).

5. Global representatives on a small polymer

The construction in this section follows [Cur, §12.7]. The scaled fields belong to the fine domain, but this alone does not give an analytic gain for their vacuum and ordinary quadratic kernels. On a small polymer they admit a stronger representation, with a real gauge, two constant fields and global fields small enough for an analytic disk of radius $L^{7/4}$.

Proposition 5.1. Assume (1). Let X be small and let $(\mathscr{A},\mathscr{C}) \in \widetilde{R}_{k+1} \times \mathfrak{X}_{k+1}$. There are a real function λ_*, constant bond functions $\mathscr{K}_1, \mathscr{K}_2$, and global bond functions $\mathscr{A}^{\mathrm{b}}, \mathscr{C}^{\mathrm{b}}$ on the fine torus such that, on every bond with an endpoint at distance at most 1 from X,

$$\mathscr{A}_L - d\lambda_* = \mathscr{A}^{\mathrm{b}} + \mathscr{K}_1, \qquad \mathscr{C}_L = \mathscr{C}^{\mathrm{b}} + \mathscr{K}_2. \tag{35}$$

They satisfy

$$\begin{aligned} \|\mathscr{A}^{\mathrm{b}}\|_\infty &\le 64L^{-15/8+3\varepsilon/2}\theta_1, \\ \|\partial\mathscr{A}^{\mathrm{b}}\|_\infty &\le 137L^{-15/8+3\varepsilon/2}\theta_2, \\ \|\delta_\alpha\partial\mathscr{A}^{\mathrm{b}}\|_\infty &\le 1382L^{-15/8+3\varepsilon/2}\theta_3, \\ \|\mathscr{C}^{\mathrm{b}}\|_\infty < L^{-7/4}d_1, &\qquad \|\partial\mathscr{C}^{\mathrm{b}}\|_\infty < L^{-7/4}d_2, \end{aligned} \tag{36}$$

and

$$\|\mathscr{K}_1\|_\infty < b_1, \qquad \|\mathscr{K}_2\|_\infty < L^{-1+\varepsilon}d_1. \tag{37}$$

Consequently

$$(t\mathscr{A}^{\mathrm{b}}, t\mathscr{C}^{\mathrm{b}}) \in R_k \times \mathfrak{X}_k \qquad (|t| \le L^{7/4}). \tag{38}$$

The construction uses a base point of X and a decomposition and local real representative of $\mathscr{A}$, and is independent of the activity. When $\mathscr{C} = 0$, it gives $\mathscr{C}^{\mathrm{b}} = \mathscr{K}_2 = 0$. ■

Proof. Choose a lift of X and a photon site c_* in it. Set $r = 2LM \ge 50$ and $K_{\mathrm{loc}} = K(c_*, 2r+5)$, with sup balls understood through their nonwrapping lifts. Choose the coarse M-cube containing c_*/L. If $\mathscr{A} = A_0 + A_1$ is its domain decomposition, its real representative A' and gauge λ on the enlarged cube give

$$a_0 = (A_0)_L = a_0' + d\lambda_L, \qquad a_0' = (A')_L, \qquad a_1 = (A_1)_L$$

on K_{loc}. To see that these functions are defined there, the scaled enlarged cube has side $(2L+1)LM$ and center z with $|c_* - z| < LM/2$. The inequality $4LM + 6 \le L^2M$ puts $K(c_*, 2r+6)$ inside this cube. Moreover $4r + 11 < \ell$, since $\ell \ge 13L^2M$. Thus the boxes and all derivative cells used below do not wrap.

We first record the cutoff estimate with its derivative support. Put

$$f(t) = \begin{cases} 1, & |t| \le 1, \\ 1 - 2(|t|-1)^2, & 1 \le |t| \le 3/2, \\ 2(2-|t|)^2, & 3/2 \le |t| \le 2, \\ 0, & |t| \ge 2. \end{cases}$$

For a bond $(c, c+\eta_k e_\mu)$ use the product of the three functions $f((c_\nu + \eta_k\delta_{\mu\nu}/2 - c_{*,\nu})/r)$ as its cutoff multiplier. Denote this operation by cut_r. The function f is continuously differentiable, $|f'| \le 2$, and f' is Lipschitz with constant 4. If a bond function b has local norms B_0, B_1, B_α on K_{loc}, extension by zero after multiplication gives

$$\begin{aligned} \|\operatorname{cut}_r b\|_\infty &\le B_0, \\ \|\partial \operatorname{cut}_r b\|_\infty &\le B_1 + 2B_0/r, \\ \|\delta_\alpha\partial \operatorname{cut}_r b\|_\infty &\le B_\alpha + 12B_1/r + 12B_0/r^2. \end{aligned} \tag{39}$$

Indeed, the multiplier has oscillation at most $6|c-c'|/r$, first difference at most $2/r$, and first-difference oscillation at most $12|c-c'|/r^2$. Apply the product rule to the first difference. In its Hölder difference the four terms are bounded by $B_\alpha|c-c'|^\alpha$, twice $6B_1|c-c'|/r$, and $12B_0|c-c'|/r^2$. Here $|c-c'| \le 1$ implies $|c-c'| \le |c-c'|^\alpha$. If any such term is nonzero, all its cells lie in $K(c_*, 2r+5)$: a nonzero multiplier or its difference requires a site within $2r+1$, the first-difference cell adds one mesh length, and the pair adds at most 1. The same bound covers the two-step cell at $k = 0$. Thus the local estimates suffice for the global ones. The cutoff equals one on bonds with both ends in $K(c_*, r)$.

For $b = a_0'$ or a_1, write $v = c - c_*$ and define

$$V[b]_{\nu\mu} = (\partial_\nu b_\mu)(c_*), \qquad \mathrm{Lin}[b]_\mu(c) = \sum_\nu V[b]_{\nu\mu}v_\nu, \qquad \mathrm{rem}[b] = b - b(c_*) - \mathrm{Lin}[b].$$

Its first difference is $\partial_\nu b_\mu(c) - \partial_\nu b_\mu(c_*)$, and its Hölder first-difference seminorm is at most b_3. Join the two sites by at most $2r+5$ simultaneous coordinate steps of length at most 1. Their difference is therefore at most $(2r+5)b_3 \le 3rb_3$. A rectilinear path to c_* has length at most $3(2r+5)$, and the remainder vanishes at c_*. Keeping the sharper derivative bound along this path and using $r \ge 50$ gives

$$\|\operatorname{rem}[b]\|_{K_{\mathrm{loc}}} \le 3(2r+5)^2 b_3 \le 18r^2 b_3, \qquad \|\partial \operatorname{rem}[b]\|_{K_{\mathrm{loc}}} \le 3rb_3.$$

Equation (39) consequently bounds the three global norms of its cutoff by $18r^2b_3$, $39rb_3$, and $253b_3$, respectively.

Only a real gauge is available. We remove the constant and symmetric linear part of a_0', leaving its curvature and the complex linear part of a_1. Set

$$V^0 = V[a_0'], \qquad V^{\mathrm{s}} = (V^0 + (V^0)^T)/2, \qquad F = V^0 - (V^0)^T,$$

and use the real quadratic function

$$q_*(c) = \sum_\mu \left[a'_{0,\mu}(c_*) - \frac{\eta_k}{2} V^{\mathrm{s}}_{\mu\mu} \right] v_\mu + \frac{1}{2} \sum_{\nu,\mu} V^{\mathrm{s}}_{\nu\mu} v_\nu v_\mu. \tag{40}$$

Its finite difference is exactly $(dq_*)_\mu(c) = a'_{0,\mu}(c_*) + \sum_\nu V^{\mathrm{s}}_{\nu\mu} v_\nu$. The mesh term in (40) cancels the diagonal term from differencing the quadratic polynomial. With $\mathscr{K}_1 = a_1(c_*)$ and

$$P_\mu(c) = \sum_\nu \left[\frac{1}{2} F_{\nu\mu} + V[a_1]_{\nu\mu} \right] v_\nu,$$

we have the local identity

$$\mathscr{A}_L - d(\lambda_L + q_*) = \mathscr{K}_1 + P + \operatorname{rem}[a_0'] + \operatorname{rem}[a_1]. \tag{41}$$

To cut off P we need a bound that also uses the side ℓ. A global mean-zero periodic function whose oscillation between sites at distance at most 1 is at most B has sup norm at most ℓB. Any two sites can be joined by at most $\lceil \ell/2 \rceil \le \ell$ simultaneous coordinate steps of length at most 1; comparison with the mean gives the claim. Every $\partial_\nu a_{1,\mu}$ has mean zero, sup norm at most b_2, and this oscillation at most b_3. For a_0, the global curl agrees on each derivative cell with the curl of a local real representative. Its sup norm is at most $2b_2$, its oscillation at most $2b_3$, and it has mean zero. The cells of a pair at distance at most 1 lie in one enlarged cube, so that the same representative can be used for their comparison. It follows that every coefficient of P is bounded by

$$s = 2\min(b_2, \ell b_3). \tag{42}$$

Use a second radius

$$r' = \min(e_k^{-\varepsilon}, \ell/5).$$

The charge condition in (1) gives $r' \ge r$, and $4r' + 11 < \ell$. Thus its cutoff is also defined without wrapping. The three norms of P on its support are at most $6.3r's$, s, and 0. Equation (39), with r' in place of r, bounds the cutoff norms by $6.3r's$, $13.6s$, and $87.6s/r'$. If $r' = e_k^{-\varepsilon}$, use $s \le 2b_2$; in units of $(\theta_1, \theta_2, \theta_3)$ these are at most $(12.6, 27.2, 175.2)L^{-15/8+\varepsilon}$. If $r' = \ell/5 < e_k^{-\varepsilon}$, use $s \le 2\ell b_3$. Since $\ell e_k^{\varepsilon} < 5$, the corresponding three bounds are at most $(63, 136, 876)L^{-15/8+3\varepsilon/2}$.

Define the global field and real gauge by

$$\begin{aligned} \mathscr{A}^{\mathrm{b}} &= \operatorname{cut}_r \left(\operatorname{rem}[a_0'] + \operatorname{rem}[a_1] \right) + \operatorname{cut}_{r'} P, \\ \lambda_* &= \lambda_L + q_* \quad \text{on } K_{\mathrm{loc}}, \qquad \lambda_* = 0 \quad \text{elsewhere.} \end{aligned} \tag{43}$$

The two remainder cutoffs contribute $(36r^2, 78r, 506)b_3$. As $r = 2LM$, the charge condition $160L^{1+3\varepsilon/2}Me_k^{\varepsilon} \le 1$ bounds the first two by $L^{-15/8+3\varepsilon/2}\theta_1$ and $L^{-15/8+3\varepsilon/2}\theta_2$, respectively. Adding the linear contribution gives the constants 64, 137, and 1382 in (36).

In the external argument take

$$\mathscr{K}_2 = \mathscr{C}_L(c_*), \qquad \mathscr{C}^{\mathrm{b}} = \operatorname{cut}_r(\mathscr{C}_L - \mathscr{K}_2).$$

A rectilinear path to c_* bounds the undifferentiated field on K_{loc} by $6.3r\|\partial\mathscr{C}_L\|_\infty$. The cutoff bounds therefore give

$$\|\mathscr{C}^{\mathrm{b}}\|_\infty \le 6.3r\|\partial\mathscr{C}_L\|_\infty, \qquad \|\partial\mathscr{C}^{\mathrm{b}}\|_\infty \le 13.6\|\partial\mathscr{C}_L\|_\infty.$$

Use (28) and $d_2 = e_k^{\varepsilon} d_1$. The first charge condition involving 160 gives the desired bound for the sup norm; the large-L condition gives $13.6L^{-2+3\varepsilon/2} \le L^{-7/4}$ for the derivative. This proves the external bounds and the constant bounds.

The diameter estimate in Appendix B puts X in $K(c_*, M(L+2))$. Every bond with an endpoint within distance 1 of X has both ends in $K(c_*, r)$, since $M(L+2)+2 \le 2LM$. All cutoffs in the local identity are one there, and the gauge is $\lambda_L + q_*$ there. This proves (35). Finally, the large-L condition implies $1382L^{-15/8+3\varepsilon/2} < L^{-7/4}$. Multiplication by any $|t| \le L^{7/4}$ consequently leaves all five norms strictly below their domain thresholds, proving (38). □

Proposition 5.2. Let $F(X)$ be a level-k Grassmann activity holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, with the kernel support and coefficient locality specified in §2.2, and covariant under the independent real gauges of its two connection arguments. For the fields of Proposition 5.1, its kernels satisfy

$$|F_{nm}(X; \mathscr{A}_L, \mathscr{C}_L; \xi; \zeta)| \le \exp(2(n+m))|F_{nm}(X; \mathscr{A}^{\mathrm{b}}, \mathscr{C}^{\mathrm{b}}; \xi; \zeta)|. \tag{44}$$

For the vacuum kernel the two values are equal. ■

Proof. For each kernel argument a let $c(a)$ be the photon site of its ordinary leg, or the first photon site of its pair leg, and let $\omega(a) \in \{0,1\}$ distinguish unbarred and barred legs. On kernels the gauge homomorphism has the form

$$(G_\lambda F)_{nm}(\xi;\zeta) = F_{nm}(\xi;\zeta) \prod_a \exp\{(-1)^{\omega(a)} i e_k \lambda(c(a))\}. \tag{45}$$

This follows by applying the homomorphism to the kernel expansion; the multiplying product is symmetric in each set of arguments, and hence preserves their antisymmetry. The gauge relation (35) first gives

$$G_{\lambda_*} F(X; \mathscr{A}^{\mathrm{b}} + \mathscr{K}_1, \mathscr{C}^{\mathrm{b}} + \mathscr{K}_2) = F(X; \mathscr{A}_L, \mathscr{C}_L).$$

Both pairs lie in the domains: a constant changes only the sup norm, and $L^{-7/4} + L^{-7/8+\varepsilon/2} < 1$ and $L^{-7/4} + L^{-1+\varepsilon} < 1$. Every phase from this real gauge has modulus one.

We remove the constant fields by continuation of a real gauge identity. Write $\mathscr{K}_j = K_j^{\mathrm{R}} + iK_j^{\mathrm{I}}$ with real constant fields. On the connected domain

$$\Omega = \{z \in \mathbb{C}^4 : |\operatorname{Im} z_1| < 2,\ |\operatorname{Im} z_2| < 2,\ |z_3| < 2,\ |z_4| < 2\},$$

set

$$A(z) = \mathscr{A}^{\mathrm{b}} + z_1 K_1^{\mathrm{R}} + z_2 K_1^{\mathrm{I}}, \qquad C(z) = \mathscr{C}^{\mathrm{b}} + z_3 K_2^{\mathrm{R}} + z_4 K_2^{\mathrm{I}},$$

and set

$$\Lambda_z(c) = -(z_1 K_1^{\mathrm{R}} + z_2 K_1^{\mathrm{I}} + z_3 K_2^{\mathrm{R}} + z_4 K_2^{\mathrm{I}}) \cdot (c - c_*) \quad (c \in K(c_*, r)),$$

with value zero elsewhere. The first connection belongs to $\widetilde{R}_k$: its complex small summand has sup norm below $(L^{-7/4} + 4L^{-7/8+\varepsilon/2})\theta_1 < \theta_1$ and derivatives those of $\mathscr{A}^{\mathrm{b}}$, and its remaining constant summand is real. Invariance under real constant additions handles its unrestricted real part. The external connection has sup norm below $(L^{-7/4} + 4L^{-1+\varepsilon})d_1 < d_1$, and derivative that of $\mathscr{C}^{\mathrm{b}}$. The large-L condition implies these two strict inequalities.

For fixed kernel arguments the function

$$H(z) = F_{nm}(X; A(z), C(z); \xi; \zeta) \prod_a \exp\{(-1)^{\omega(a)} i e_k \Lambda_z(c(a))\}$$

is holomorphic on Ω. When z is real, split Λ_z into its first two and last two terms. These are independent real gauge functions for the two connections, with differentials equal to the negatives of their respective constants on $K(c_*, r)$. Every bond read by the local gauge condition lies in this ball.

The real gauge identity and (45) therefore give $H(z) = H(0)$ for $z \in \mathbb{R}^2 \times (-2,2)^2$. All derivatives of $H - H(0)$ at zero vanish, so its holomorphic power series is zero there. The identity theorem extends this equality to the connected domain Ω. Evaluating at $z = (1, i, 1, i)$ gives the identity for the two original complex constants.

It remains to bound the inverse phases. By locality, an ordinary argument lies over X. For a pair of the first kind its first photon site also lies in X. For a boundary pair that site has distance at most $\eta_k \le 1$ from X. Hence in all nonzero kernel terms

$$|c(a) - c_*|_1 \le 3(M(L+2)+1) \le 9LM.$$

The constant bounds and the charge conditions give

$$e_k |K_1^{\mathrm{I}} \cdot (c(a) - c_*)| \le 9LML^{-7/8+\varepsilon/2} e_k^{1/4+\varepsilon} \le 1,$$
$$e_k |K_2^{\mathrm{I}} \cdot (c(a) - c_*)| \le 9ML^{\varepsilon} e_k^{2\varepsilon} \le 1.$$

Each leg phase, and its inverse, thus has modulus at most $\exp(2)$. Combining this bound with the unit modulus phases from λ_* proves (44). The vacuum has no leg phases, which proves its equality. □

6. Contraction of the normalized activity

On a small polymer the ordinary quadratic kernel now vanishes at zero connection. The vacuum is also even in the connection, so its first derivative vanishes there. The analytic disk gives an additional gain for precisely these two terms; every other term is controlled directly by its Grassmann degree.

Proposition 6.1. Assume (21), (1), and (2). For $\mathscr{E} \in K_k^{x,\mathbb{C}}(\ell)$ set $E = (R\mathscr{E})^{\mathrm{nat}}$. The linear map $\mathscr{E} \mapsto \mathcal{L}E$ takes $K_k^{x,\mathbb{C}}(\ell)$ into $K_{k+1}^{x,\mathbb{C}}(\ell/L)$, preserves the real subspaces, and satisfies

$$\|\mathcal{L}E\|_{k+1}^x \le c_{\mathrm{fl}} L^{-1/4+3\varepsilon/2} \|\mathscr{E}\|_k^x \le c_{\mathrm{fl}} L^{-1/4+2\varepsilon} \|\mathscr{E}\|_k^x, \qquad c_{\mathrm{fl}} = 61 c_{\mathrm{rb}}. \tag{46}$$

For the weights $w_j^{\mathrm{E}} = e_j^{1/4-7\varepsilon}$,

$$\frac{\|\mathcal{L}E\|_{k+1}^x}{w_{k+1}^{\mathrm{E}}} \le c_{\mathrm{fl}} L^{-3/8+11\varepsilon/2} \frac{\|\mathscr{E}\|_k^x}{w_k^{\mathrm{E}}}. \tag{47}$$

Restriction to the zero external argument commutes with the map. ■

Proof. Write

$$N(X) = \sup_{\widetilde{R}_k \times \mathfrak{X}_k} \|E(X;\cdot,\cdot)\|_{h_k}.$$

Extraction and adjustment give

$$N(X) \le \begin{cases} 20\|\mathscr{E}\|_k^x \exp(-\kappa d_M(X)), & X \text{ small}, \\ \|\mathscr{E}\|_k^x \exp(-\kappa d_M(X)), & X \text{ large}. \end{cases} \tag{48}$$

For a small polymer this uses $\|R\mathscr{E}(X;\mathscr{A},\mathscr{C})\|_{h_k} \le \|\mathscr{E}(X;\mathscr{A},\mathscr{C})\|_{h_k} + \|\mathscr{E}(X;0,0)\|_{h_k}$ and (25). For a large polymer both operations leave the activity unchanged.

Fix a small X and a coarse connection pair, and use the global representatives above. Put $\rho = L^{7/4}$. The inverse image of the open set $R_k \times \mathfrak{X}_k$ under $t \mapsto (t\mathscr{A}^{\mathrm{b}}, t\mathscr{C}^{\mathrm{b}})$ contains the closed disk $|t| \le \rho$, hence a slightly larger disk. The functions

$$f_{nm}(t) = E_{nm}(X; t\mathscr{A}^{\mathrm{b}}, t\mathscr{C}^{\mathrm{b}})$$

are holomorphic there, with values in a finite-dimensional kernel space. Their weighted norms obey

$$\frac{h_{k,1}^n h_{k,2}^m}{n!m!} \|f_{nm}(t)\|_1 \le N(X) \qquad (|t| \le \rho).$$

Here $\|\cdot\|_1$ is the sum of kernel moduli with their index weights.

For the vacuum, (24) gives $f_{00}(0)=0$. Charge conjugation fixes a scalar kernel and reverses both connections, so $f_{00}(-t)=f_{00}(t)$ and $f_{00}'(0)=0$. Cauchy's formula gives

$$f_{00}(1)=\frac{1}{2\pi i}\oint_{|t|=\rho}\frac{f_{00}(t)}{t^2(t-1)}\,\mathrm{d}t.$$

The vacuum equality in Proposition 5.2 and $\rho\ge 2$ therefore give

$$\|(S_LE)^{[0,0]}(X;\mathscr{A}_L,\mathscr{C}_L)\|_{h_{k+1}}\le\frac{N(X)}{\rho(\rho-1)}\le 2L^{-7/2}N(X). \tag{49}$$

For the ordinary quadratic kernel, normalization gives $f_{20}(0)=0$ as a full antisymmetric kernel. Its balanced matrix components vanish by (24); their oppositely ordered components vanish by antisymmetry, and the two equally barred or equally unbarred components vanish by balance. Hence

$$f_{20}(1)=\frac{1}{2\pi i}\oint_{|t|=\rho}\frac{f_{20}(t)}{t(t-1)}\,\mathrm{d}t,\qquad \|f_{20}(1)\|_1\le\frac{\sup_{|t|=\rho}\|f_{20}(t)\|_1}{\rho-1}.$$

The comparison of kernels costs $\exp(4)$ in this bidegree. Its scaling exponent is $p(2,0)=9/4$. Multiplying the last inequality by $h_{k,1}^2/2$ and using (32) gives

$$\|(S_LE)^{[2,0]}(X;\mathscr{A}_L,\mathscr{C}_L)\|_{h_{k+1}}\le 2\exp(4)L^{-4}N(X). \tag{50}$$

For all remaining bidegrees use the original scaled connection pair, which belongs to the fine domain. Evenness leaves only $(1,1)$, $(0,2)$, and total degrees at least 4. The exponents satisfy

$$p(1,1)>\frac{13}{4}-\frac{3\varepsilon}{2},\qquad p(0,2)>\frac{13}{4}-\frac{3\varepsilon}{2},\qquad p(n,m)\ge n+m.$$

Consequently

$$\begin{aligned}\sum_{(n,m)\neq(0,0),(2,0)}\|(S_LE)^{[n,m]}(X;\mathscr{A}_L,\mathscr{C}_L)\|_{h_{k+1}}&\le L^{-13/4+3\varepsilon/2}\sum_{n,m}\|E^{[n,m]}(X;\mathscr{A}_L,\mathscr{C}_L)\|_{h_k}\\&\le L^{-13/4+3\varepsilon/2}N(X).\end{aligned}$$

The sum is taken at one connection pair before bounding it by $N(X)$. The large-L condition ensures $2L^{-7/2}\le L^{-13/4}$ and $2\exp(4)L^{-4}\le L^{-13/4}$. Adding the three degree contributions yields

$$\|S_LE(X;\mathscr{A}_L,\mathscr{C}_L)\|_{h_{k+1}}\le 3L^{-13/4+3\varepsilon/2}N(X). \tag{51}$$

We now sum over the fine polymers in a coarse polymer Y. The hypotheses (2) imply $\kappa-1\ge\kappa_0$ and $L(\kappa-\kappa_0-2)\ge\kappa$. The reblocking bound (83) at rate κ and (48) therefore give

$$\sum_{\substack{X\text{ small}\\ \bar{X}^L=LY}}\|S_LE(X;\mathscr{A}_L,\mathscr{C}_L)\|_{h_{k+1}}\le 60c_{\mathrm{rb}}L^{-1/4+3\varepsilon/2}\|\mathscr{E}\|_k^x\exp(-\kappa d_M(Y)).$$

For large X, scaling decreases the norm by (31) and the ratios of the two weights. Since $d_M(X)>L$,

$$\exp(-\kappa d_M(X))\le\exp(-L)\exp(-(\kappa-1)d_M(X)).$$

Use (83) at rate $\kappa-1$ to obtain

$$\sum_{\substack{X\text{ large}\\ \bar{X}^L=LY}}\|S_LE(X;\mathscr{A}_L,\mathscr{C}_L)\|_{h_{k+1}}\le c_{\mathrm{rb}}L^3\exp(-L)\|\mathscr{E}\|_k^x\exp(-\kappa d_M(Y)).$$

As $L^3\exp(-L)\le L^{-1/4}$, these estimates prove (46) after taking the two suprema in the activity norm. Division by $w_{k+1}^{\mathrm{E}}=L^{1/8-7\varepsilon/2}w_k^{\mathrm{E}}$ gives the relative factor $c_{\text{fl}}L^{-3/8+5\varepsilon}$, which is at most the factor in (47). The additional parameter choice $c_{\text{fl}}L^{-3/8+6\bar{\varepsilon}}\le 1/16$ makes its relative factor at most $1/16$.

The map is linear. Its kernels are finite sums of holomorphic fine kernels composed with the complex linear connection scaling. The locality was proved after (34). For lattice covariance a coarse group element $g' = (S, a')$ has the fine representative $g = (S, La')$. It gives a bijection of the polymers in the two reblocking sums, satisfies $(g'\mathscr{A})_L = g\mathscr{A}_L$, and intertwines the spin actions by $a_{g'}S_L = S_L a_g$. The same identity holds for $\mathscr{C}$. Thus covariance of each fine activity gives covariance of its sum. For the real gauges, every fine bond within distance 1 of X scales to a bond within distance $1/L \le 1$ of Y, while $G_\lambda S_L = S_L G_{\lambda_L}$ follows from $e_k\lambda_L(c) = e_{k+1}\lambda(c/L)$. Independent real gauges of the two arguments therefore pass through each term. Finally, scaling has real coefficients, preserves the bar index, and commutes with charge conjugation and the antilinear reality map. These facts prove membership in both stated activity spaces. Since extraction and adjustment commute with restriction and $0_L = 0$, reblocking also commutes with restriction to the zero external argument. □

The reblocked sum retains the evaluated normalized activity. Indeed, each fine polymer belongs to exactly one reblocking class, so

$$\sum_Y (\mathcal{L}E)(Y; \mathscr{A}, \mathscr{C}) = S_L \sum_X E(X; \mathscr{A}_L, \mathscr{C}_L). \tag{52}$$

If the coarse generators are evaluated at fields with $\chi' = \delta_{\alpha,\mathscr{A}+\mathscr{C}}\psi'$ and its barred counterpart, their scaled fields satisfy the fine relations. This follows from $d_X(x/L, x'/L) = d_X(x, x')/L$ and $U_{(\mathscr{A}+\mathscr{C})_L}(x, x') = U_{\mathscr{A}+\mathscr{C}}(x/L, x'/L)$: the path bijection and $e_k\eta_k L^{-1/2} = e_{k+1}\eta_{k+1}$ give the latter. Substitution in the difference quotient produces $L^{-1-\alpha}\chi'(x/L, x'/L)$, and the barred quotient uses the inverse transport. Adjustment consequently has its original value under this evaluation. Thus the evaluation of (52) is also the evaluation of

$$S_L \sum_X (R\mathscr{E})(X; \mathscr{A}_L, \mathscr{C}_L).$$

Proof of Theorem A. Proposition 3.1 gives the scalar counterterms and their bounds. Equations (18) and (19) give the exact extraction and the integrated normalization. The shortest-chain construction (22) preserves evaluated values; (24) and (25) give the full kernel normalization and its bound. Proposition 6.1 proves the contraction and the relative estimate. The covariance and restriction statements were proved with each map. □

7. Local reconstruction of a mean field

The result in this section follows [Cur, §§12.4, 13.4] and has hypotheses independent of the large-L contraction conditions. Fix $k \ge 1$, $\eta_k = L^{-k}$, $L \ge 5$ odd, $M = L^m$ with $m \in \mathbb{Z}_{\ge 2}$, and an admissible torus as in Section 2. Let $0 < \alpha < 1$, $0 < \rho < 1$, $0 < \mu \le 1$, and $b_k \ge b_- > 0$.

The chamber Dirac stencil is

$$D_A\psi(x) = -\eta_k^{-1} \sum_{i=0}^{3} M_i(x)[U_A(x, x^{(i)})\psi(x^{(i)}) - \psi(x)], \qquad M_i(x) = \tfrac{1}{2}w_i(\rho - \gamma \cdot \widehat{d}_i(x)),$$

where

$$w_0 = w_3 = 6, \qquad w_1 = w_2 = 6\sqrt{2},$$
$$\widehat{d}_0 = -ge_3, \quad \widehat{d}_1 = -g(e_2 - e_3)/\sqrt{2}, \quad \widehat{d}_2 = -g(e_1 - e_2)/\sqrt{2}, \quad \widehat{d}_3 = ge_1.$$

In particular $\sum_i w_i\widehat{d}_i = 0$ and $\|M_3(x)\|_2 \le 6$.

Let Δ_y be the unit block centered at $y \in \mathbb{Z}^3/\ell\mathbb{Z}^3$. For $c \in \Delta_y$, join c to the successive centers of its nested blocks of sides $\eta_k, L\eta_k, \dots, 1$. Let $\varphi_A(y, c)$ be the exponential of $ie_k\eta_k$ times the sum of

the six-path averages along these successive center-to-site paths, as in (58). Define the normalized average and its reverse by

$$(\mathcal{Q}_k(A)v)(y) = \frac{\eta_k^3}{48}\sum_{c\in\Delta_y,g}\varphi_A(y,c)v(c,g), \qquad (\mathcal{Q}_k^\sharp(A)u)(c,g) = \varphi_A(y,c)^{-1}u(y). \tag{53}$$

Thus $\mathcal{Q}_k\mathcal{Q}_k^\sharp = 1$. Equivalently $\mathcal{Q}_k^\sharp(A) = \mathcal{Q}_k(-A)^T$, where T is the algebraic transpose with the stated fine measure and unit counting measure. Reversing the connection and taking this weighted transpose are separate operations. Put

$$F_\infty(A) = D_A + \mu + b_k\mathcal{Q}_k^\sharp(A)\mathcal{Q}_k(A), \qquad T_k(A) = \mathcal{Q}_k(A)(b_k^{-1}D_A + 1 + \mu/b_k).$$

Write $\bar{T}_k(A) = \mathsf{C}^{-1}T_k(-A)\mathsf{C}$. For a complex connection set $\mathfrak{i}(A) = e_k\|\operatorname{Im}A\|_\infty$ and $q_d(A) = e_k\|dA\|_\infty$. For a scalar-component kernel $P(a,b)$ acting with column measure $\nu(b)$, write

$$|P|_{1,\infty} = \sup_a\sum_b \nu(b)|P(a,b)|.$$

The sum includes spin indices. For ordinary and pair columns the measures are $\eta_k^3/48$ and $(\eta_k^3/48)^2$, respectively.

Theorem B (Local reconstruction)**.** Under the preceding hypotheses the following statements hold.

(i) For every complex connection A and every union Y of partition M-cubes, the rows $T_Y^{(1)}(A)$ and $T_Y^{(2)}(A)$ in (61) exist. The first uses only ordinary legs over Y; the second uses the external ordered pairs leaving Y. Their coefficients read only bonds meeting Y, are entire in A, and satisfy

$$T_Y^{(1)}(A)\psi + T_Y^{(2)}(A)\delta_{\alpha,A}\psi = T_k(A)\psi \quad \text{at the unit sites in } Y.$$

They obey the lattice, real-gauge and reality covariance identities (67)–(69). Their barred counterparts are $\bar{T}_Y^{(i)}(A) = \mathsf{C}^{-1}T_Y^{(i)}(-A)\mathsf{C}$.

(ii) In Euclidean spinor supremum norm, writing $\mathfrak{i} = \mathfrak{i}(A)$, $q_d = q_d(A)$,

$$\begin{aligned}\|T_Y^{(1)}(A)\|_{\mathrm{su}},\ \|T_k(A)\|_{\mathrm{su}} &\le e^{3\mathfrak{i}/2}[1 + b_-^{-1}(54Lq_de^{6q_d} + 12 + 6e^{\mathfrak{i}} + \mu)],\\ \|T_Y^{(2)}(A)\|_{\mathrm{su}} &\le 6b_-^{-1}e^{3\mathfrak{i}/2}\eta_k^\alpha.\end{aligned} \tag{54}$$

If $\mathfrak{i}, q_d \le 1$, the coefficient row sums satisfy

$$|T_Y^{(i)}(A)|_{1,\infty},\ |T_k(A)|_{1,\infty},\ |\bar{T}_Y^{(i)}(A)|_{1,\infty},\ |\bar{T}_k(A)|_{1,\infty} \le C_T \quad (i = 1,2), \tag{55}$$

where

$$C_T = 2e^{3/2}[1 + b_-^{-1}(54Le^6 + 12e + 1)].$$

This constant is independent of k, ℓ, Y. In particular the bound holds for $A \in \widetilde{R}_k + \mathfrak{X}_k$ when $0 < \varepsilon \le 1/200$, $0 < e_k < 1$, and (8) holds.

(iii) Introduce independent unit generators $\Psi, \bar{\Psi}$ over Y. Replacing them by $T_Y^{(1)}\psi + T_Y^{(2)}\chi$ and its barred counterpart, and fixing the old generators, defines the homomorphism $\mathrm{sub}_{Y,A}$ of (70). It preserves balance and the symmetry identities (71). If the original ordinary legs lie over Y and its pair legs are admissible for Y, the substituted polynomial has the same support properties. If $\mathfrak{i}, q_d \le 1$ and the ordinary, pair and unit weights $w_1, w_2, w_3 > 0$ satisfy $C_T(w_1 + w_2) \le w_3$, then

$$\|\operatorname{sub}_{Y,A}F\|_{w_1,w_2} \le \|F\|_{w_1,w_2;w_3}$$

for every polynomial whose unit generators lie over Y.

(iv) At each A for which $F_\infty(A)$ is invertible, define

$$H_k(A) = b_k F_\infty(A)^{-1} \mathcal{Q}_k^\sharp(A).$$

Then $T_k(A)H_k(A) = 1$. For the barred recovery also assume $F_\infty(-A)$ is invertible and define $\bar{H}_k(A) = \mathsf{C}^{-1} H_k(-A)\mathsf{C}$. On evaluation at $\psi = H_k(A)\Phi$, $\chi = \delta_{\alpha,A}\psi$, and the barred analogues, the substitution sends $\Psi, \bar{\Psi}$ to $\Phi, \bar{\Phi}$ over Y.

■□

The inverse hypothesis in the last clause is additional to the row estimate. The proof below constructs the rows directly. Reindexing the internal chamber hops cancels the bulk part of the Dirac stencil. The remaining boundary rows are controlled by face counts, while the averaging-phase defect is controlled by curvature and the first common block scale.

7.1. **The rows and their cancellation.** The normalized average (53) gives

$$T_k(A) = b_k^{-1} \mathcal{Q}_k(A)(D_A + \mu + b_k \mathcal{Q}_k^\sharp(A)\mathcal{Q}_k(A)) = \mathcal{Q}_k(A)(b_k^{-1} D_A + 1 + \mu/b_k). \tag{56}$$

The second form follows from $\mathcal{Q}_k \mathcal{Q}_k^\sharp = 1$. At a connection where $F_\infty(A)$ is invertible, its mean map $H_k(A) = b_k F_\infty(A)^{-1} \mathcal{Q}_k^\sharp(A)$ satisfies

$$T_k(A)H_k(A) = b_k^{-1} \mathcal{Q}_k(A) F_\infty(A) b_k F_\infty(A)^{-1} \mathcal{Q}_k^\sharp(A) = 1. \tag{57}$$

The barred identity follows where the inverse at $-A$ exists, by conjugation with C.

For $c \in \Delta_y$, let $c_j(c)$ be the center of its block of side $L^j \eta_k$, $0 \le j \le k$, so that $c_0(c) = c$ and $c_k(c) = y$. The averaging phase is

$$\varphi_A(y, c) = \exp\left(i e_k \eta_k \sum_{j=0}^{k-1} (\tau A)(c_{j+1}(c), c_j(c)) \right). \tag{58}$$

The sum uses the average of the six rectilinear path sums. We write $(\tau_k A)(y, c)$ for this sum. All of these paths lie in Δ_y and their total length in bonds is at most $\frac{3}{2}(L^k - 1)$. Thus

$$\begin{aligned} |\varphi_A(y, c)^{\pm 1}| &\le \exp(3\mathfrak{i}(A)/2), \\ |u_A(c, c')^{\pm 1}| &\le \exp(\eta_k \mathfrak{i}(A)) \le \exp(\mathfrak{i}(A)). \end{aligned} \tag{59}$$

for a photon bond, with $u_A(c, c') = \exp(i e_k \eta_k A(c, c'))$.

Write $c_3 = \pi x^{(3)} = c + \eta_k g e_1$. The row form of (56) is

$$\begin{aligned} (T_k(A)\psi)(y) = \int_{\Delta_y} \mathrm{d}x\, \varphi_A(y, c) \Big\{ &(1 + \mu/b_k)\psi(x) \\ &- (b_k \eta_k)^{-1} \sum_{i=0}^{3} M_i(x)[U_A(x, x^{(i)})\psi(x^{(i)}) - \psi(x)] \Big\}. \end{aligned} \tag{60}$$

Here $U_A = 1$ for $i \le 2$ and $U_A(x, x^{(3)}) = u_A(c, c_3)$. The normalized chamber integral over Δ_y has total mass one. This row exists for every complex connection, regardless of invertibility; the inverse hypothesis was needed only for (57).

The rows map antiperiodic fields to antiperiodic coarse fields: translation by a period preserves blocks, neighbors and phases, and supplies the single antiperiodic sign from the source field.

If Y is a union of M-cubes and $y \in Y$ is a unit site, then $\Delta_y \subset Y$: both M and L^k are odd, and the unit block is contained in the M-cube of its center. Define the two rows

$$\begin{aligned}(T_Y^{(1)}(A)\psi)(y) &= \int_{\Delta_y} \mathrm{d}x\, \varphi_A(y,c)\Big\{(1+\mu/b_k)\psi(x) \\ &\qquad - (b_k\eta_k)^{-1} \sum_{i:\, x^{(i)} \text{ over } Y} M_i(x)[U_A(x,x^{(i)})\psi(x^{(i)}) - \psi(x)]\Big\}, \\ (T_Y^{(2)}(A)\chi)(y) &= -b_k^{-1}\eta_k^{\alpha-1}\int_{\Delta_y} \mathrm{d}x\, 1[c_3 \notin Y]\varphi_A(y,c)M_3(x)\chi(x,x^{(3)}).\end{aligned} \tag{61}$$

Rows outside Y are zero. The first row contains ordinary legs only over Y. The second row contains precisely the type-3 pairs leaving Y, which are admissible boundary pairs of length η_k. Their coefficients read A only on bonds with an endpoint in Δ_y. If $\chi = \delta_{\alpha,A}\psi$ on these boundary pairs, then

$$T_Y^{(1)}(A)\psi + T_Y^{(2)}(A)\chi = T_k(A)\psi \quad \text{on the unit sites of } Y. \tag{62}$$

This follows by replacing the missing type-3 difference in (60) by $\eta_k^\alpha \chi(x,x^{(3)})$. The intra-cluster differences remain in $T_Y^{(1)}$.

Proposition 7.1 (Uniform row bounds)**.** The rows (61) and $T_k(A)$ obey the Euclidean norm estimates (54) for every complex connection A. If $\mathfrak{i}(A), q_d(A) \le 1$, their coefficient row sums obey (55), with the stated constant C_T independent of k, ℓ, Y. ∎

Proof. Write $\mathfrak{i} = \mathfrak{i}(A)$ and $q_d = q_d(A)$. Put $E = \Delta_y \times O_{\mathrm{cub}}$ and divide it into

$$E_1 = \{x \in E : c_3 \in \Delta_y\}, \quad E_2 = \{x \in E : c_3 \in Y \setminus \Delta_y\}, \quad E_3 = \{x \in E : c_3 \notin Y\}.$$

For each coordinate direction $a \in \{\pm e_\mu\}$ there are eight labels with $ge_1 = a$. The face of Δ_y through which $c + \eta_k a$ leaves the unit block has η_k^{-2} photon sites. Thus

$$\#E = 48\eta_k^{-3}, \qquad \#(E_2 \cup E_3) = 48\eta_k^{-2}. \tag{63}$$

The chamber hopping matrices satisfy

$$M_i(x^{(i)}) - M_i(x) = w_i\gamma\cdot\widehat{d}_i(x), \qquad \sum_{i=0}^{3} w_i\gamma\cdot\widehat{d}_i(x) = 0.$$

Also $\|M_3(x)\|_2 \le 6$ and $\|w_3\gamma\cdot\widehat{d}_3(x)\|_2 = 6$. Reindex each intra-cluster sum by $x \mapsto x^{(i)}$ and each type-3 sum in E_1 by $x \mapsto x^{(3)}$. Both are involutions of their summation sets. The intra-cluster sums become $\sum_{x\in E}\varphi_A(y,c)\sum_{i\le 2}(M_i(x^{(i)}) - M_i(x))\psi(x)$, which equals $-\sum_{x\in E}\varphi_A(y,c)w_3\gamma\cdot\widehat{d}_3(x)\psi(x)$. The part over E_1 cancels with the corresponding matrix difference from the reindexed type-3 sum. Thus $(T_Y^{(1)}(A)\psi)(y) = R_0 + R_1 + R_2 + R_3$, with

$$\begin{aligned}R_0 &= (1+\mu/b_k)\frac{\eta_k^3}{48}\sum_{x\in E}\varphi_A(y,c)\psi(x), \\ R_1 &= b_k^{-1}\frac{\eta_k^2}{48}\sum_{x\in E_2\cup E_3}\varphi_A(y,c)w_3\gamma\cdot\widehat{d}_3(x)\psi(x), \\ R_2 &= -b_k^{-1}\frac{\eta_k^2}{48}\sum_{x\in E_2}\varphi_A(y,c)M_3(x)[u_A(c,c_3)\psi(x^{(3)}) - \psi(x)], \\ R_3 &= -b_k^{-1}\frac{\eta_k^2}{48}\sum_{x\in E_1}[\varphi_A(y,c_3)u_A(c_3,c) - \varphi_A(y,c)]M_3(x^{(3)})\psi(x).\end{aligned} \tag{64}$$

The face count and phase moduli bound the first three terms by

$$\exp(3\mathfrak{i}/2)\|\psi\|_{\mathrm{su}}[1 + \mu/b_k + 6b_k^{-1} + 6b_k^{-1}(1+\exp(\mathfrak{i}))].$$

The fourth term requires the curvature estimate.

For neighboring photon sites $c, c' = c + \eta_k a$ in Δ_y, let $j(c,a)$ be the first scale at which their block centers agree. The centers at all larger scales then agree as well. For $j \geq 2$, such a c lies on a face of a block of side $L^{j-1}\eta_k$. There are $L^{3(k-j+1)}$ such blocks inside Δ_y, with $L^{2(j-1)}$ sites on the indicated face. For $j = 1$ use all sites. We obtain

$$\#\{c : c, c + \eta_k a \in \Delta_y,\ j(c,a) = j\} \leq \eta_k^{-3} L^{-(j-1)}. \tag{65}$$

Write

$$\begin{aligned}\varphi_A(y,c')u_A(c',c) - \varphi_A(y,c) &= \varphi_A(y,c)(\exp(v(c,a)) - 1),\\ v(c,a) &= ie_k\eta_k[(\tau_k A)(y,c') + A(c',c) - (\tau_k A)(y,c)].\end{aligned}$$

The contributions above scale $j(c,a)$ cancel in this bracket. The remaining expression is the mean of the line sums over closed paths from their common scale-j center down to c', then across the bond to c, and back up the reversed paths. All these loops lie in the scale-j block, of side at most $L^j\eta_k$, and have at most $3L^j$ bonds. For a closed path of n bonds in a box of side s, the axial path estimate gives $|A(\Gamma)| \leq 2ns\|dA\|_\infty$. To recall the reason, subtract the gauge obtained by line sums from one corner in coordinate order $1, 2, 3$. The difference on each bond is a sum over at most two strips of width one mesh and length at most s. Its modulus is at most $2s\|dA\|_\infty$, and the gauge sum around the loop vanishes. This argument applies to complex fields as well. Consequently, with $j = j(c,a)$,

$$|v(c,a)| \leq 6L^{2(j-k)}q_d, \qquad |\exp(v(c,a)) - 1| \leq 6L^{2(j-k)}q_d \exp(6q_d). \tag{66}$$

The factor $L^{2(j-k)}$ must be kept until after summing the locations of the first common block. By (65),

$$\begin{aligned}\eta_k^2 \sum_{\substack{c\in\Delta_y\\ c+\eta_k a\in\Delta_y}} |\exp(v(c,a)) - 1| &\leq 6q_d \exp(6q_d) L^{1-k} \sum_{j=1}^{k} L^j\\ &\leq 9Lq_d \exp(6q_d).\end{aligned}$$

The six directions, eight labels per direction, and $\|M_3\|_2 \leq 6$ now give

$$|R_3|_2 \leq 54Lq_d \exp(6q_d) b_k^{-1} \exp(3\mathfrak{i}/2)\|\psi\|_{\mathrm{su}}.$$

Together with the first three terms this proves the first line of (54) for $T_Y^{(1)}$. Take Y to be the full torus to obtain the bound for T_k.

The second row is supported on E_3. Its definition and (63) give directly

$$|(T_Y^{(2)}(A)\chi)(y)|_2 \leq b_k^{-1}\eta_k^{\alpha-1}\frac{\eta_k^3}{48}48\eta_k^{-2}\exp(3\mathfrak{i}/2)6\|\chi\|_{\mathrm{su}},$$

which is the second bound.

For the coefficient norm, fix a row and choose each source component to have the conjugate phase of its row coefficient. Its component supremum is at most one, and its Euclidean spinor supremum is at most two. The row coefficient sum is then at most twice the corresponding bound in (54). When $\mathfrak{i}, q_d \leq 1$, use also $\mu \leq 1$ and $\eta_k^\alpha \leq 1$. The resulting bound is C_T as stated. The normalized index measure is already in the row coefficients, so no extra factor 48 enters. The domain is invariant under sign, and the matrix C is monomial; conjugating $T_Y^{(i)}(-A)$ or $T_k(-A)$ by it permutes row and column moduli. This proves the barred bounds. For a total connection in $\widetilde{R}_k + \mathfrak{X}_k$, (10) supplies $\mathfrak{i}, q_d \leq 1$ under (8). □

The local reconstruction respects the symmetries of an activity. Telescoping the scalar gauge sum in the averaging paths gives

$$\varphi_{A-d\lambda}(y,c)\exp(ie_k\lambda(c)) = \exp(ie_k\lambda(y))\varphi_A(y,c),$$
$$u_{A-d\lambda}(c,c')\exp(ie_k\lambda(c')) = \exp(ie_k\lambda(c))u_A(c,c').$$

Insert these two identities in (61). Every term acquires the coarse factor $\exp(ie_k\lambda(y))$, where the boundary leg transforms with the phase of its first point. Thus, with λ^{u} the restriction to unit sites,

$$\begin{aligned} T_Y^{(1)}(A-d\lambda)\exp(ie_k\lambda)\psi &= \exp(ie_k\lambda^{\mathrm{u}})T_Y^{(1)}(A)\psi, \\ T_Y^{(2)}(A-d\lambda)\chi^\lambda = \exp(ie_k\lambda^{\mathrm{u}})T_Y^{(2)}(A)\chi, &\qquad \chi^\lambda(x,x') = \exp(ie_k\lambda(c))\chi(x,x'). \end{aligned} \tag{67}$$

The identity also holds for gauges specified locally on the bonds and sites read by the row.

A scheme symmetry $\tilde{g} = (S,a)$ maps nested blocks and their centers to their images, preserves the order of each coordinate path, and permutes the 48 labels. Moreover $M_i(\tilde{g}x)S = SM_i(x)$ and $(\tilde{g}x)^{(i)} = \tilde{g}(x^{(i)})$. Changing variables in the two rows therefore proves

$$T_{\tilde{g}Y}^{(i)}(\tilde{g}A)\tilde{g} = \tilde{g}T_Y^{(i)}(A). \tag{68}$$

Complex conjugation changes the phases to those at $-A$, and $\overline{M_i} = \varkappa^{-1}M_i\varkappa$, giving

$$\overline{T_Y^{(i)}(\overline{A})} = \varkappa^{-1}T_Y^{(i)}(-A)\varkappa. \tag{69}$$

The barred rows are $\bar{T}_Y^{(i)}(A) = \mathsf{C}^{-1}T_Y^{(i)}(-A)\mathsf{C}$. Their gauges have the opposite sign. For the lattice covariance, $(S^{-1})^T = \sigma(S)\mathsf{C}^{-1}S\mathsf{C}$ with $\sigma(S)\in\{\pm1\}$: both matrix maps are homomorphisms, and on each generating Dirac matrix they differ by a minus sign. This same sign occurs in the source and target spin actions and cancels in the barred intertwining identity. For reality, C is real and $\varkappa\mathsf{C} = -\mathsf{C}\varkappa$; the two signs from commuting the conjugations cancel. Thus the barred versions of (68) and (69) hold with the inverse transpose spin action and the same $\varkappa$.

7.2. **Local substitution of the mean field.** For $k\ge1$, let Y be a union of partition M-cubes and allow independent unit generators $\Psi,\bar{\Psi}$ over Y, together with the ordinary and Hölder generators. Using the two operators from (61), define the substitution that fixes the latter generators and sets, at the unit sites of Y,

$$\begin{aligned} \mathrm{sub}_{Y,B}\,\Psi &= T_Y^{(1)}(B)\psi + T_Y^{(2)}(B)\chi, \\ \mathrm{sub}_{Y,B}\,\bar{\Psi} &= \bar{T}_Y^{(1)}(B)\bar{\psi} + \bar{T}_Y^{(2)}(B)\bar{\chi}. \end{aligned} \tag{70}$$

The images are odd linear combinations of independent generators and therefore anticommute with one another and with the fixed generators. They determine a unique Grassmann homomorphism. Its new ordinary legs lie over Y and its new pair legs are boundary pairs of the second kind with first point over Y. Its coefficients read B on bonds meeting Y, by the row support proved in Section 7. Since a bar index is preserved, the substitution preserves balance and parity.

If an evaluation has $\chi = \delta_{\alpha,B}\psi$ and the barred relation on these boundary pairs, then (62) turns (70) into $T_k(B)\psi$ and $\bar{T}_k(B)\bar{\psi}$. If in addition $\psi = H_k(B)\Phi$ and $\bar{\psi} = \bar{H}_k(B)\bar{\Phi}$, the left-inverse identity gives Φ and $\bar{\Phi}$ over Y. Thus this local substitution represents the exact mean field under evaluation.

The row covariances yield

$$\begin{aligned} a_g\,\mathrm{sub}_{gY,gB} &= \mathrm{sub}_{Y,B}\,a_g, \\ G_\lambda\,\mathrm{sub}_{Y,B^1} &= \mathrm{sub}_{Y,B}\,G_\lambda, \qquad B^1 = B - d\lambda \text{ on bonds meeting } Y, \\ \mathrm{Cg}\,\mathrm{sub}_{Y,-B} &= \mathrm{sub}_{Y,B}\,\mathrm{Cg}, \\ \mathbf{K}\,\mathrm{sub}_{Y,-\overline{B}} &= \mathrm{sub}_{Y,B}\,\mathbf{K}. \end{aligned} \tag{71}$$

The final relation uses real μ. Both sides fix the old generators according to the stated symmetry, so it suffices to compare their images of $\Psi, \bar{\Psi}$. The lattice and real-gauge relations follow from (68) and (67); for a local gauge equality the rows read only the bonds on which that equality is assumed. For charge conjugation use $T_Y^{(i)}(-B)\mathsf{C} = \mathsf{C}\bar{T}_Y^{(i)}(B)$ and its barred counterpart. For the antilinear map, (69) with $-B$ gives $\overline{T_Y^{(i)}(-\overline{B})} = \varkappa^{-1}T_Y^{(i)}(B)\varkappa$, and likewise for the barred row. These identities give (71) on each new generator.

Assume $\mathfrak{i}(B), q_d(B) \le 1$ under the standing hypotheses of this section. Proposition 7.1 gives the following norm statement. Assign weights w_1, w_2 to the ordinary and pair legs and weight w_3 to the independent unit generators:

$$\|\operatorname{sub}_{Y,B} F\|_{w_1,w_2} \le \|F\|_{w_1,w_2;w_3} \quad \text{if } C_T(w_1+w_2) \le w_3. \tag{72}$$

Indeed the image of a Ψ generator has norm at most $C_T w_1 + C_T w_2$, and the original generators retain their weights. The weighted linear-substitution bound in Lemma 2.1 applies on the subalgebra whose unit generators lie over Y. The measures for the two row sums are $\eta_k^3/48$ and $(\eta_k^3/48)^2$, respectively; their identity kernels include the reciprocal column measure. Thus this estimate has the same normalization as the reconstruction bound.

Proof of Theorem B. Equations (56)–(62) give the row construction, its locality and the evaluated identity. Proposition 7.1 proves the quantitative estimates; its proof uses only $\mathfrak{i}, q_d \le 1$ for the displayed constant C_T. The generator checks and norm argument in Section 7.2 give the substitution assertions. Where the indicated inverses exist, (57) gives recovery. □

Appendix A. Lifted formulas and comparison of local data

The extraction, adjustment and scaling maps of Sections 3 and 4 have the following local formulas, as in [Cur, §12.9]. For comparison of different torus volumes we need formulas that depend on a lift and local data alone. The required margin also distinguishes the pairs produced by scaling from the full set of coarse pairs.

For a finite photon set $\widetilde{U}$ on the covering lattice, let $\mathfrak{G}_k(\widetilde{U})$ be the algebra generated by the ordinary indices over $\widetilde{U}$ and pair indices whose two photon sites lie in $\widetilde{U}$. Its ordinary and pair measures remain $\eta_k^3/48$ and $(\eta_k^3/48)^2$. For a system of separately antisymmetric kernels $K = (K_{nm})$ define

$$\mathrm{El}_k(K) = \sum_{n,m} \frac{1}{n!m!} \int \mathrm{d}\xi\, \mathrm{d}\zeta\, K_{nm}(\xi;\zeta)\psi(\xi_1)\cdots\psi(\xi_n)\chi(\zeta_1)\cdots\chi(\zeta_m).$$

The integrals range over these two finite index sets. The kernel expansion is a bijection between the systems of such kernels and this algebra: at increasing arguments its coefficient is the kernel times the positive product of index weights, as in (4).

Map each covering generator to the torus generator at its simultaneous period class, with the antiperiodic convention, and denote the homomorphism by cov_ℓ. If

$$\operatorname{diam}_\infty(\widetilde{U}) + \eta_k < \ell, \tag{73}$$

it is injective and preserves kernels when read on this lift. In fact two congruent photon sites in $\widetilde{U}$ must coincide. Choose a fundamental domain containing $\widetilde{U}$ and order its independent ordinary and pair indices so that they extend the covering orders. The homomorphism maps distinct covering basis monomials to distinct torus basis monomials with the same coefficients and weights. This proves injectivity, preservation of bidegrees, and equality of the kernels at the lifted arguments. All torus kernels at indices outside these period classes vanish.

Now let $\widetilde{X}$ be a face-connected lift and put $\widetilde{U} = \widetilde{X}^{+\eta_k}$, its mesh neighborhood. The margin

$$\operatorname{diam}_\infty(\widetilde{X}) + 2\eta_k + M < \ell \tag{74}$$

gives a bijection of ordinary indices over $\widetilde{X}$ onto those over X, and of admissible pairs for $\widetilde{X}$ onto those for X. For a pair of the first kind, its endpoints and every point of its shortest paths lift inside $\widetilde{X}$: a lifted path point within distance M of its first endpoint differs from its representative in $\widetilde{X}$ by a period of norm at most $M + \operatorname{diam}_\infty(\widetilde{X}) < \ell$. That period is zero. For a boundary pair $x' = x^{(3)}$, lift the endpoint in X; the other is its prescribed nearest neighbor in $\widetilde{U}$. Two lifted pairs in the same class coincide because their first photon sites coincide. These arguments prove surjectivity, injectivity and preservation of pair kind. The margin also implies (73) for $\widetilde{U}$. By the nonwrapping lifting result, X is small exactly when $d_M(\widetilde{X}) \le L$: for either a small torus polymer or a small covering polymer the unique nonwrapping lift has the same net length.

Take data $(K^0, K, \widetilde{B})$ consisting of two kernel systems over $\widetilde{U}$ and a connection on bonds with both ends in $\widetilde{X}$. For $d_M(\widetilde{X}) \le L$, define

$$\begin{aligned} M^0 &= \operatorname{Vol}(\widetilde{X})^{-1} \int_{\widetilde{X}} \mathrm{d}x' \int_{\widetilde{X}} \mathrm{d}x''\, K^0_{20}(x', x''), \\ \Phi^R &= \operatorname{El}_k(K) - K^0_{00} - \int_{\widetilde{X}} \mathrm{d}x\, \bar\psi(x)^T M^0 \psi(x), \\ \Phi^E &= \Phi^R - (\Phi^R)^{[2,0]}_{\widetilde{X}} + \Theta_{\widetilde{X},\widetilde{B}} (\Phi^R)^{[2,0]}_{\widetilde{X}}. \end{aligned} \tag{75}$$

Here K^0_{20} is the balanced matrix component in the earlier convention, and the subscript $\widetilde{X}$ retains only quadratic terms whose two sites lie over $\widetilde{X}$. Thus the transport homomorphism is applied within its domain. For $d_M(\widetilde{X}) > L$, set $\Phi^R = \Phi^E = \operatorname{El}_k(K)$. In both cases define

$$\Phi^S = \widetilde{S}_L \Phi^E,$$

where $\widetilde{S}_L$ is the covering version of (30). These formulas use no side ℓ. Translation by $v \in M\mathbb{Z}^3$ preserves the cube geometry, measures and chain transports. Therefore

$$\Phi^R(T_v \text{data}) = T_v \Phi^R(\text{data}), \qquad \Phi^E(T_v \text{data}) = T_v \Phi^E(\text{data}), \qquad \Phi^S(T_v \text{data}) = T_{v/L} \Phi^S(\text{data}), \tag{76}$$

with the translated polymer understood on the left.

For an activity $\mathscr{E}$, let K^0 and K be its kernels at $(0,0)$ and at $(\mathscr{A}_L, \mathscr{C}_L)$, restricted to the lift, and let $\widetilde{B}$ be the restriction of $(\mathscr{A} + \mathscr{C})_L$ to the internal bonds. Then

$$\begin{aligned} (R\mathscr{E})(X; \mathscr{A}_L, \mathscr{C}_L) &= \operatorname{cov}_\ell \Phi^R, \\ (R\mathscr{E})^{\mathrm{nat}}(X; \mathscr{A}_L, \mathscr{C}_L) &= \operatorname{cov}_\ell \Phi^E, \\ S_L[(R\mathscr{E})^{\mathrm{nat}}(X; \mathscr{A}_L, \mathscr{C}_L)] &= \operatorname{cov}_{\ell/L} \Phi^S. \end{aligned} \tag{77}$$

To prove the first identity, locality and the index bijection reproduce the original activity from K. The vacuum subtraction is exactly K^0_{00}, and the two label-averaged mass integrals have the same weights and volume in the lift and on the torus. The diagonal bilinear is period invariant, since its two antiperiodic signs cancel. The subtraction is therefore also the same. For the second identity the ordinary quadratic terms of Φ^R lie over $\widetilde{X}$. Cubes, their shortest chains and their transports are the lifted versions of the torus construction. On each ordinary generator this gives $\operatorname{cov}_\ell \Theta_{\widetilde{X},\widetilde{B}} = \Theta_{X,(\mathscr{A}+\mathscr{C})_L} \operatorname{cov}_\ell$. The homomorphism identity proves the adjusted formula. Finally, on each generator

$$\operatorname{cov}_{\ell/L} \widetilde{S}_L = S_L \operatorname{cov}_\ell .$$

Both sides have the same scaled index and coefficient, with compatible antiperiodic signs. This proves the third identity.

The margin (74) suffices for the image of $\widetilde{S}_L$, whose pairs have $d_X \le M/L$. To identify $\operatorname{cov}_{\ell/L}$ with the quotient map on the full coarse local algebra, including pairs with $M/L < d_X \le M$, we

use the stronger margin

$$\operatorname{diam}_\infty(\widetilde{U}) + LM < \ell. \tag{78}$$

It is equivalent to $\operatorname{diam}_\infty(\widetilde{U}/L) + M < \ell/L$. For a small polymer both margins hold, since

$$\operatorname{diam}_\infty(\widetilde{X}) + 2\eta_k + LM \le 2LM + 4M < \ell.$$

The torus expressions in (77) do not depend on the chosen lift. Translation of that lift by ℓw translates the data; their total even degree makes the product of their argument signs one. Equation (76) and the same evenness then cancel the generator signs under the quotient maps at both levels. Thus identical lifted data produce identical lifted extraction, adjustment and scaled kernels on different admissible tori, subject to the stated margins.

Appendix B. Polymer geometry

We give the geometric estimates used above, following [Cur, Appendix A].

Let L be odd, M a power of L, and ℓ a multiple of LM with $\ell \ge 13LM$. The M-cubes have centres in $M\mathbb{Z}^3/\ell\mathbb{Z}^3$. They partition the lattice sites; their closed continuum versions may meet along faces and edges. A polymer is a nonempty union of M-cubes connected through faces. Denote its number of cubes by $|X|_M$. A net in the continuum torus is a connected finite union of images of specified closed straight segments in $\mathbb{R}^3$, a point being allowed as a segment. Its length is the sum of the Euclidean lengths of those segments. Set

$$M d_M(X) = \inf\{\text{length}(N) : N \text{ is a net meeting every closed cube of } X\},$$

where the infimum also extends over the finite segment representations. The same definition is used for finite face-connected unions of M-cubes in $\mathbb{R}^3$, with nets in $\mathbb{R}^3$. Distances are sup distances.

B.1. **Size, diameter, and reblocking.** Put

$$c_X = 54, \qquad \kappa_0 = 54\log 72, \qquad K_0 = e^{\kappa_0}/36, \qquad c_{\mathrm{rb}} = c_X K_0.$$

The following estimates are uniform in M and ℓ:

$$d_M(X) \le |X|_M - 1, \qquad |X|_M \le c_X(d_M(X)+1), \tag{79}$$

$$d(p,q) \le M(d_M(X)+2) \qquad (p,q \text{ in the closed cubes of } X), \tag{80}$$

$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \le K_0, \qquad \sum_{X \subset Y} e^{-\kappa_0 d_M(X)} \le K_0 |Y|_M. \tag{81}$$

Here Y is any union of M-cubes. If ℓ/L satisfies the same side conditions stated above, let $\bar{X}^L$ be the union of LM-cubes containing cubes of X. There is a unique polymer Y on the rescaled torus with $\bar{X}^L = LY$, and

$$L d_M(Y) \le d_M(X), \tag{82}$$

$$\sum_{X:\bar{X}^L = LY} e^{-\kappa' d_M(X)} \le c_{\mathrm{rb}} L^3 e^{-L(\kappa'-\kappa_0-1)d_M(Y)} \quad (\kappa' \ge \kappa_0). \tag{83}$$

To prove the first inequality of (79), order the cubes along a spanning tree of their adjacency graph. A net meeting one cube can be extended to an adjacent cube by a segment of length at most M perpendicular to their common face. Starting with one point and repeating this construction gives a net of length at most $M(|X|_M - 1)$.

For the reverse inequality, subdivide the segments of any net N at their intersections and discard overlapping pieces. Its total length does not increase. Doubling the edges of the resulting connected graph gives a closed traversal of length at most $2\,\text{length}(N)$. Cut it into at most $2\,\text{length}(N)/M + 1$ arcs of length at most M. Each arc lifts to a curve in $\mathbb{R}^3$ whose range in each coordinate has length

at most M. Such an interval meets at most three closed cube intervals, so an arc meets at most 27 closed M-cubes. Every cube of X meets one of these arcs. Taking the infimum over N gives

$$|X|_M \le 27(2d_M(X)+1) \le c_X(d_M(X)+1).$$

For (80), take a net of length less than $Md_M(X)+\delta$ and points p',q' of the net in the respective cubes of p,q. The path in the net from p' to q' has length at most that of the net. Hence $d(p,q) \le M + (Md_M(X)+\delta) + M$; let $\delta \downarrow 0$. The same argument holds in $\mathbb{R}^3$.

A polymer with n cubes containing a fixed cube determines a closed walk of length $2(n-1)$ starting there: double a spanning tree of its adjacency graph and traverse it. Choose one such walk for each polymer. This assignment is injective, since the visited cubes recover the polymer, and there are at most $6^{2(n-1)} = 36^{n-1}$ possible walks. By the size estimate, $d_M(X) \ge n/c_X - 1$. Summing over n gives

$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \le e^{\kappa_0} \sum_{n \ge 1} 36^{n-1} 72^{-n} = K_0.$$

Sum this bound over the cubes of Y to obtain the second inequality of (81).

Every M-cube belongs to one LM-cube, since L is odd, and adjacent M-cubes belong to the same or to adjacent LM-cubes. Thus $\bar{X}^L$ is connected through faces. Projection by $c \mapsto c/L$ gives its unique rescaled polymer Y. A net for X meets every LM-cube of $\bar{X}^L$; its rescaling has length divided by L. This proves (82). For (83), split the decay into rates κ_0 and $\kappa' - \kappa_0$ and use (82). All terms satisfy $X \subset LY$, which has $L^3|Y|_M$ fine cubes, so

$$\sum_{X : \bar{X}^L = LY} e^{-\kappa_0 d_M(X)} \le K_0 L^3 |Y|_M \le c_X K_0 L^3 e^{d_M(Y)}.$$

Combining the two factors proves the stated bound.

B.2. **Nonwrapping lifts.** Let $\mathrm{pr} : \mathbb{R}^3 \to \mathbb{R}^3/\ell\mathbb{Z}^3$ be the projection.

Proposition B.1. If X is a polymer and

$$2M(d_M(X)+2) < \ell, \tag{84}$$

then $X = \mathrm{pr}(\widetilde{X})$ for a finite face-connected union of M-cubes in $\mathbb{R}^3$. The projection is injective on $\widetilde{X}$, preserves adjacency of its cubes in both directions, and $d_M(\widetilde{X}) = d_M(X)$. The lift is unique up to a translation in $\ell\mathbb{Z}^3$. ■

Proof. Put $D = M(d_M(X)+2)$, fix p_0 in a closed cube of X, and fix its lift $\widetilde{p}_0$. The projection is injective on the closed sup ball $\widetilde{B} = \{|p - \widetilde{p}_0|_\infty \le D\}$, since two points there differ in every coordinate by at most $2D < \ell$. It maps this ball onto the torus ball $B = \{p : d(p,p_0) \le D\}$. Its inverse $\varphi : B \to \widetilde{B}$ is continuous, and is locally isometric. The diameter estimate places every closed cube of X in B. A segment or a closed M-cube lying in B lifts under φ to a segment of the same length or a closed M-cube: the translates $\widetilde{B} + \ell n$ are pairwise disjoint, and a connected lift lies in one of them. The same inverse maps shared faces to shared faces. Thus $\widetilde{X} = \varphi(X)$ has the asserted injectivity, adjacency, and connectedness.

Any other face-connected injective lift can be translated so that it contains the chosen lift of the cube of p_0. There are no two different neighbours of one lifted cube with the same projection: their centres differ by at most $2M < \ell$. Propagate this observation along a chain of adjacent cubes. Every cube of the other lift equals the corresponding cube of $\widetilde{X}$, proving uniqueness.

Projecting a net for $\widetilde{X}$ gives a net for X with no greater length, so $d_M(X) \le d_M(\widetilde{X})$. Conversely, let N be a net for X of length at most $Md_M(X) + M$, and choose its point q_0 in the cube of p_0. Every point of N is joined to q_0 by a curve of length at most that of N, and $d(q_0,p_0) \le M$. Therefore $N \subset B$. Its lift $\varphi(N)$ has the same length and meets every cube of $\widetilde{X}$. Taking an infimizing sequence of these nets proves the reverse inequality. □

References

[Cur] “The field-strength Osterwalder–Schrader axioms for QED in three dimensions”. Comprehensive edition, 263 pp., 20 October 2026, before the modular integration revision. 2026 (cit. on pp. 2, 3, 6, 10, 11, 17, 23, 25).